SGTX Platform

Master Blueprint

Clean Master Edition v14.0 — FULLY EXPANDED

Sovereign Governed Trade Execution Infrastructure

STATUS: AUDITED / INTEGRATED / CANONICAL / FULLY EXPANDED

Document Date: 2026-08-26

Prepared by: Master Blueprint Integration Engine

Classification: Internal Technical Master Specification

Sources: v13.1 FINAL (47,216 lines) + v12.0 (76,122 lines) + Change-Set (7,582 lines)

Non-Custodial | Non-Marketplace | Governor-Governed | USTN-Centric | Jurisdiction-Aware

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Executive Summary — What Changed from v13.1 and Why

This Clean Master Edition v14.0 is the fully expanded, institutionally-defensible master blueprint for the SGTX platform. It consolidates the v13.1 FINAL document (47,216 lines of source text), the v12.0 baseline (76,122 lines), and the complete change-set (7,582 lines) into a single, coherent, three-layer specification. Every principle, every architecture component, every portal workflow, and every implementation artefact is documented here with full traceability to its source.

The v13.1 baseline is treated as the audited historical record. All changes in v14.0 are additive, clarifying, or restructuring. No material, principle, or capability has been silently deleted. The 24 audit findings from v13.1 Part A (A-01 through A-24) are enforced as governing language — settled conflicts are not re-opened.

Primary structural change: The document is separated into three strict layers. Layer 0 (Constitution) contains immutable principles only — amendable via multisig + notice process. Layer 1 (Architecture) contains the current normative architecture — mutable through standard change-management. Layer 2 (Implementation Specifications) contains concrete, testable artefacts — canonical data models, event catalogues, API contracts, and observability requirements. Every normative statement is tagged with its layer: [L0], [L1], or [L2].

Over-claim elimination: All residual marketing language has been replaced with precise deployment-state vocabulary. 'Production-ready' is replaced with CORE_READY or PRODUCTION_CONNECTED. 'Complete' is replaced with 'specified' or 'implemented'. 'Zero-cost' is replaced with 'institutional-cost scope clarified' (data sources are free; institutional costs — bank fees, broker fees, integration costs — are real). This vocabulary change is not cosmetic — it reflects the constitutional requirement that SGTX never over-claims its capabilities to banks, regulators, or institutional users.

Portal documentation: This edition adds comprehensive portal-by-portal documentation (Part V) covering all 12 portals. Each portal's tabs, screens, workflows, and data access patterns are documented in full. This is the primary addition over v13.1 — the v13.1 document described portal architecture but did not document each portal's complete user-facing surface.

Trade lifecycle documentation: The 36-stage end-to-end trade workflow (blueprint Article 129) is documented in full (Part VI), with each stage's inputs, outputs, authority class, and portal touch-points specified. This enables an engineering team to implement the complete lifecycle without ambiguity.

What v14.0 is NOT: It is not a new architecture. It does not introduce new capabilities beyond what v13.1 specified. It is a clean, restructured, fully-expanded presentation of the same architecture, with all contradictions resolved, all over-claims eliminated, all principles clearly separated by mutability layer, and all portal workflows documented end-to-end.

How to Read This Document (Layer System)

This document uses a three-layer tagging system. Every normative statement is prefixed with its layer:

Layer	Tag	Mutability	Content	Amendment
Layer 0	[L0]	Immutable	Constitutional principles (G1-G7, 32-point constitution, AI ladder, non-custody, non-marketplace)	Multisig 3-of-5 + 30-day notice
Layer 1	[L1]	Mutable (standard change-mgmt)	Normative architecture (state vector, event spine, Governor, settlement, add-ons, transport engines)	Standard change-management process (Appendix C.3)
Layer 2	[L2]	Mutable (engineering)	Implementation artefacts (data models, event catalogues, API contracts, observability)	Engineering change request + code review

Precedence rule: When a conflict exists between layers, the lower layer prevails. Layer 0 overrides Layer 1 overrides Layer 2. No implementation decision (L2) may violate an architectural principle (L1), and no architectural decision (L1) may violate a constitutional principle (L0).

Deployment-state vocabulary: Throughout this document, the following precise terms are used instead of imprecise marketing language:

Term	Meaning	Replaces
CORE_READY	Implemented, tested, passes adversarial test suite	'production-ready', 'complete'
ADAPTER_READY	Connector code exists but no live integration	'integrated'
COUNTRY_CONFIGURED	Jurisdiction profile loaded but not connected	'supported'
SANDBOX_CONNECTED	Sandbox integration active, production not yet	'connected'
PRODUCTION_CONNECTED	Live production integration active with all 4 readiness dimensions	'live'
LEGAL_AUTHORIZATION_REQUIRED	Technical ready, awaiting legal/regulatory approval	'pending'
MANUAL_ONLY	Process requires manual operation, no API	'supported manually'
PORTAL_ONLY	Process via government portal, no API integration	'web-based'
INTEGRATION_REQUIRED	Integration not yet built, required for full operation	'missing'

PART II — LAYER 0: CONSTITUTION

[L0] This layer contains immutable principles only. These principles are the non-negotiable foundation of the SGTX platform. They may be amended only via an explicit multisig + notice process (see §0.5). No implementation decision, business pressure, or operational convenience may override a Layer 0 principle. Every component, every integration, and every trade on the SGTX platform must comply with all Layer 0 principles at all times.

0.1 Governor Principles (G1–G7)

[L0] The Governor is the constitutional enforcement engine of SGTX. Seven principles govern its operation. These principles are immutable — they cannot be waived, overridden, or bypassed by any component, any user, or any configuration.

Principle	Statement	Enforcement Mechanism
G1 — Execution Always Gated	Every irreversible action requires Governor approval before execution. No component may bypass the Governor pipeline.	Every API endpoint that performs a state mutation calls governorDecide() before executing. The Governor's verdict (ALLOW/CONDITIONAL/DENY) is recorded in the Loom.
G2 — OPA Enforced	Open Policy Agent (OPA) evaluates every decision against authored policies. Policy violations block execution.	OPA is called as a sidecar in the Governor pipeline. If OPA returns DENY, the Governor returns DENY regardless of other gates.
G3 — WasmEdge Constitutional	WasmEdge executes constitutional rules (non-custody, non-marketplace, 110% reserve, closure-is-earned) as deterministic WebAssembly modules.	WasmEdge modules are compiled from the Layer 0 principles. They cannot be overridden by configuration, policy, or human action.
G4 — Loom Audited	Every Governor decision is appended to the Loom (SHA-256 hash-chained audit log). The Loom is immutable and externally verifiable.	The Loom uses a SHA-256 hash chain where each entry's hash is computed over (previousHash + payload + timestamp). Any modification breaks all subsequent hashes.
G5 — Multisig for Irreversible	Irreversible actions (USTN closure, policy amendment, fund release) require multisig approval.	Standard: 2-of-3 for trade-level irreversible actions. Constitutional: 3-of-5 for platform-level changes. Signatories use QES (Qualified Electronic Signature).
G6 — AI Advisory Only	The AI subsystem (A1–A3) may propose, explain, classify, and escalate, but NEVER has independent execution authority.	A4 automation is deterministic policy execution by Governor+OPA+WasmEdge, NOT AI autonomy. The AI subsystem cannot call any state-mutating API directly.
G7 — Bank-Authoritative Settlement	SGTX orchestrates settlement instructions but never becomes the settlement authority. Banks and regulated financial institutions confirm settlement finality.	The Bank Settlement Gateway sends instructions to banks but never confirms settlement. Settlement confirmation is an Event emitted by the bank integration, not by SGTX.

0.2 The 32-Point SGTX Transaction Constitution

[L0] The following 32 points constitute the immutable constitution of the SGTX platform. Every trade, every component, and every integration must comply with all 32 points. Violation of any point is a constitutional breach that triggers a SEV-0 incident and automatic Governor block.

[L0] Point 1: Non-custodial — SGTX never holds customer funds. FeeLock escrow is non-custodial; funds remain with the regulated bank/PSP until settlement is confirmed. The platform's architecture makes it structurally impossible for SGTX to receive, hold, or transfer customer funds at any point in the trade lifecycle.

[L0] Point 2: Non-marketplace — SGTX does not match buyers with sellers. Trades occur between known, relationship-controlled counterparties. The platform provides no public listing, no public ranking, no public recommendation, and no autonomous provider selection. Counterparty relationships are explicitly established (approved, connected, saved GTID).

[L0] Point 3: Non-title-taking — SGTX never takes title to goods. Title transfer is governed by the contract Incoterm and applicable law. The platform records title transfer events but never becomes the title holder or intermediary in the title chain.

[L0] Point 4: Non-carrier — SGTX is not a carrier. It orchestrates logistics execution through approved carriers, LSPs, and shipping lines. The platform does not own vehicles, vessels, aircraft, or rolling stock, and does not employ drivers, pilots, or crew.

[L0] Point 5: Non-customs-authority — SGTX is not a customs authority. It interfaces with customs authorities (Nafeza, single-window systems) but never replaces them. The platform submits declarations on behalf of brokers but never issues customs clearance — clearance is an authoritative act by the customs authority.

[L0] Point 6: Non-bank — SGTX is not a bank. It orchestrates payment instructions but never holds deposits, executes settlement, or issues credit. The platform does not have a banking licence and does not perform any regulated banking activity.

[L0] Point 7: Non-deposit-taking — SGTX does not accept deposits. All funds flow through connected banks and regulated PSPs. The platform's FeeLock mechanism locks the platform fee amount but does not collect or hold the fee — collection happens at settlement by the bank.

[L0] Point 8: Non-government — SGTX is not a government entity. It serves as infrastructure connecting commercial parties with government systems. The platform does not issue regulations, impose tariffs, or grant licences — it integrates with government systems that perform these functions.

[L0] Point 9: AI-assisted — AI provides advisory (A1), constraining (A2), and escalation (A3) capabilities. A4 automation is deterministic policy execution by the Governor. The AI subsystem enhances human decision-making but never replaces it for authoritative acts.

[L0] Point 10: Governor-governed — Every irreversible action passes through the Governor pipeline (G1–G7). No component may bypass the Governor. The Governor's decision is final — if it returns DENY, no other component may override it.

[L0] Point 11: OPA-enforced — Open Policy Agent evaluates every decision against authored policies. Policies are versioned, reviewed, and deployed through the change-management process. Policy violations are blocking — there is no 'override' mechanism for OPA denials.

[L0] Point 12: WasmEdge-enforced — Constitutional rules execute as deterministic WebAssembly modules. These modules are compiled from Layer 0 principles and cannot be overridden by configuration, policy, or human action. They are the cryptographic guarantee that the constitution is enforced.

[L0] Point 13: Loom-audited — Every decision is appended to the SHA-256 hash-chained Loom audit log. The Loom is immutable (append-only), externally verifiable, and retained for the regulatory retention period (7 years for trade data). No entry is ever modified or deleted.

[L0] Point 14: USTN-centric — The Universal Shipment Tracking Number (USTN) is the canonical namespace for every trade. All downstream records — shipments, customs declarations, payments, documents, inspections — reference the USTN. The USTN is minted at contract lock and remains the immutable reference for the trade's entire lifecycle.

[L0] Point 15: Jurisdiction-aware — Every trade is evaluated against the applicable jurisdiction's regulatory profile, customs procedure, and legal framework. The jurisdiction profile determines which licences, permits, certificates, and government integrations are required. Trades that cross jurisdictions are evaluated against all applicable jurisdictions.

[L0] Point 16: Relationship-controlled — Counterparty relationships are explicitly established (approved, connected, saved GTID). No random matching, no public rankings, no unsolicited recommendations. A provider can become available to a trader only because: (1) it is already approved/connected, (2) the trader saved its GTID, (3) the trader explicitly selected it, or (4) a government-mandated service relationship exists.

[L0] Point 17: 110% reserve rule — If reserve metadata is maintained, it must be at least 110% backed. The constitutional layer sets the threshold; ZK attestation (Add-On 7) provides cryptographic evidence that the threshold is met. The 110% rule is immutable — it cannot be lowered by configuration or policy.

[L0] Point 18: Closure-is-earned — A USTN is closed only when all 7 closure conditions are met (delivery accepted, settlement complete, reconciliation complete, customs complete, post-clearance complete, disputes satisfied, evidence sealed). Closure cannot be forced, skipped, or overridden. The canClose predicate is a pure function that evaluates all 7 conditions.

[L0] Point 19: Recovery ≠ erasure — Recovery actions (exception resolution, obligation failure cascade) restore state but never erase audit history. The Loom is immutable — corrections are made by appending new events that reference the original, never by modifying or deleting historical events. This ensures full audit trail integrity.

[L0] Point 20: USTN as namespace, not override — The USTN is a universal reference namespace. It does not override external authoritative systems (government references, bank transaction IDs, customs declarations). The USTN is SGTX's internal canonical reference; external systems maintain their own identifiers, which are mapped to the USTN via the External Identifier Registry.

[L0] Point 21: Bank-authoritative settlement — Banks and regulated financial institutions are authoritative for money movement and settlement confirmation. SGTX orchestrates instructions; it does not confirm settlement. Settlement confirmation is an Event emitted by the bank integration, not by SGTX. The Bank Settlement Gateway is non-custodial — it sends instructions but never holds funds.

[L0] Point 22: Non-custody is architectural — Non-custody is an architectural property of SGTX, not a standalone legal classification. Actual functionality determines licensing in each jurisdiction. A jurisdiction may classify SGTX's orchestration role as requiring specific licences regardless of the non-custody architecture. The Regulatory Classification Gate (Part 7) maps functionality to licence class.

[L0] Point 23: Reserve metadata ≠ custody — Reserve tables store attestations, ratios, commitments, and evidence metadata only. They do not record the reserves themselves and never give SGTX control over reserve assets. The tables record proof that reserves exist and are sufficient; they do not create customer-fund custody.

[L0] Point 24: Stablecoin/DeFi conditional — Stablecoin/DeFi rails are conditional, jurisdiction-permitted financing capabilities, not canonical settlement authority. They are used only where legally permitted and always as a sub-rail beneath bank-authoritative settlement. No DeFi reference is deleted; each is conditioned on jurisdictional permission and subordination to bank-authoritative settlement.

[L0] Point 25: GNN non-marketplace bounded — The Graph Neural Network (Add-On 1) provides trust analytics for parties already known to the tenant. It may score network trust and exposure, but it may not discover, recommend, rank, or introduce counterparties. The graph answers 'how much do I already trust a party I already know'; it never answers 'which party should I trade with'.

[L0] Point 26: Direct API = first-party connector — 'Direct API' denotes a currently adopted first-party connector (e.g., Nafeza, CargoX, ETA, CBE). The worldwide adapter fabric is the extensibility layer. The four Egypt connectors are the first concrete realisations of the adapter fabric; they are not a limitation on the global model.

[L0] Point 27: RoRo is first-class — Roll-on/Roll-off cargo is a first-class transport mode, not a sub-mode of ocean container. It has its own entity types (RoRoShipment, RoRoUnit with VIN-level tracking, RoRoVoyage, RoRoYard, RoRoGateEvent, RoRoInspection, RoRoBillOfLading), its own 19-state unit state machine, its own 12-state vessel state machine, and its own terminal adapters.

[L0] Point 28: Mode-specific government applicability — Government integrations (e.g., Egypt Nafeza) have mode-specific applicability. A single generic maritime workflow is insufficient. ROAD, AIR, OCEAN_CONTAINER, RORO, and RAIL each have distinct customs procedures, government messages, identifiers, ACI requirements, manifest formats, and declaration requirements.

[L0] Point 29: External readiness is 4-dimensional — Every integration reports independently across four dimensions: TECHNICAL (API connected?), LEGAL (contracts signed?), OPERATIONAL (procedures tested?), COMMERCIAL (fees agreed?). 'Connected' requires all four dimensions to be satisfied. This prevents the false claim 'connected' when only the technical dimension is met.

[L0] Point 30: Production-readiness vocabulary — Use CORE_READY, ADAPTER_READY, COUNTRY_CONFIGURED, SANDBOX_CONNECTED, PRODUCTION_CONNECTED, LEGAL_AUTHORIZATION_REQUIRED, MANUAL_ONLY, PORTAL_ONLY, INTEGRATION_REQUIRED. Never claim WORLDWIDE_INTEGRATED without evidence. The Production Readiness Center (Government Portal tab 24) honestly reports the current state using this vocabulary.

[L0] Point 31: Manual fallback is governed — Manual fallback (API → EDI → SFTP → PORTAL → BROKER → MANUAL) is authenticated, attributable, timestamped, documented, hashed, and Loom-logged. Manual actions are never anonymous — the actor, the action, the timestamp, and the reason are recorded in the Loom for audit.

[L0] Point 32: Evidence is sealed at closure — The final evidence package (26 categories per Art 101) is sealed at USTN closure. Post-closure observation (§22) may add evidence but cannot modify sealed evidence. The sealed evidence package is the immutable audit artefact that regulators, banks, and auditors use for post-trade inspection.

0.3 AI Authority Ladder (A0–A5)

[L0] The AI subsystem operates on a strict authority ladder. A5 is FORBIDDEN — no component may implement A5 under any circumstances. The ladder is immutable and applies to every AI capability in the platform.

Level	Name	What AI May Do	What AI May NOT Do	Boundary Statement
A0	None	Nothing — pure deterministic rules	Any AI involvement	No AI subsystem is invoked; all decisions are rule-based.
A1	Advisory	Explain, translate, suggest, summarise, notify, generate draft instructions, provide context	Make decisions, block actions, force outcomes, override human judgment	AI output is advisory only. Humans may accept, modify, or reject AI suggestions freely. AI cannot block any action.
A2	Constraining	Classify, detect anomalies/discrepancies, predict delays, compare images, estimate ETA, optimise ULD/stowage, analyse route, identify risks	Make final decisions, enforce constraints autonomously, override human authority	AI proposes constraints; the Governor decides whether to enforce. AI identifies risks; humans resolve them.
A3	Escalation	Escalate to human review, flag for enhanced due diligence, trigger compliance review, request additional evidence	Resolve escalations autonomously, close escalations without human review	AI escalates; humans resolve. Every A3 escalation requires a human decision to close.
A4	Execution (within bounds)	Deterministic policy automation executed by Governor + OPA + WasmEdge under pre-authorised rules. Example: auto-approve a trade that passes all compliance gates.	AI acquires independent execution authority; AI makes decisions outside pre-authorised rules; AI overrides Governor/OPA/WasmEdge	A4 is policy execution, NOT AI autonomy. The AI subsystem proposes the action; the Governor + OPA + WasmEdge execute it deterministically. The AI never calls state-mutating APIs directly.
A5	FORBIDDEN	NOTHING	Autonomous AI decision-making without human authorization; AI that can execute trades, release funds, or modify state without human approval	CONSTITUTIONALLY PROHIBITED. No component may implement A5. Any attempt to implement A5 is a constitutional breach triggering SEV-0 incident and automatic platform shutdown.

[L0] **Key boundary:** The AI subsystem NEVER has independent execution authority. A4 is deterministic policy automation — the AI proposes, the Governor + OPA + WasmEdge execute under pre-authorised rules. The AI subsystem itself never acquires execution authority. This is the constitutional boundary that cannot be crossed.

0.4 Design Philosophy Statements

[L0] The following design philosophy statements are immutable. They guide every architectural decision and every implementation choice.

[L0] Sovereignty First — SGTX respects the sovereignty of every jurisdiction it operates in. The platform does not override, bypass, or supersede any jurisdiction's laws, regulations, or authorities. Government systems are authoritative; SGTX is the orchestration layer that connects to them.

[L0] Non-Custody by Architecture — Non-custody is not a policy choice — it is an architectural property. The platform's code makes it structurally impossible to hold customer funds. This is enforced by WasmEdge constitutional modules that block any code path that would result in SGTX receiving, holding, or transferring customer funds.

[L0] Relationship-Controlled, Not Marketplace — SGTX facilitates trade between known parties. It does not create a public marketplace. There are no public listings, no public rankings, no public recommendations. Counterparty relationships are explicitly established and controlled by the trader.

[L0] AI Assists, Never Forces — AI enhances human decision-making but never replaces it for authoritative acts. The AI subsystem can advise, constrain, and escalate, but the final decision is always made by a human (for A1-A3) or by deterministic policy (for A4). AI never forces an outcome.

[L0] Closure is Earned — A trade is not closed until all 7 closure conditions are met. There is no 'force close' mechanism. This ensures that every closed trade is fully settled, fully reconciled, and fully evidenced — providing a clean audit trail for regulators and auditors.

[L0] Recovery Does Not Erase — When something goes wrong, the platform recovers state but never erases history. The Loom audit log is immutable. Corrections are made by appending new events, not by modifying or deleting old ones. This ensures full audit trail integrity even in the face of errors and exceptions.

[L0] Evidence is Sealed — At trade closure, the evidence package is sealed — it becomes immutable. Post-closure observation may add new evidence but cannot modify sealed evidence. This ensures that the audit trail at closure is a reliable snapshot of the trade's lifecycle.

[L0] Precise Vocabulary — The platform uses precise deployment-state vocabulary (CORE_READY, PRODUCTION_CONNECTED, etc.) instead of marketing language. This prevents over-claiming and ensures that banks, regulators, and institutional users have an accurate picture of the platform's capabilities.

0.5 Amendment Process

[L0] Layer 0 principles may be amended only through the following process:

Step 1 — Proposal — A formal amendment proposal is submitted to the Platform Governance Authority. The proposal must include: the principle being amended, the new language, the rationale, and the impact assessment.

Step 2 — Notice Period — The proposal undergoes a mandatory 30-day notice period during which all affected parties (banks, regulators, traders, platform operators) may review and comment. The notice period ensures transparency and gives stakeholders time to assess impact.

Step 3 — Multisig Approval — The amendment requires 3-of-5 multisig approval from the constitutional signatories. Signatories are identified by GTID and must use Qualified Electronic Signatures (QES). The multisig ensures no single party can amend the constitution.

Step 4 — Loom Logging — The amendment is appended to the Loom with its full provenance: proposer, reviewers, signatories, timestamp, rationale, and the text of both the old and new versions. This ensures complete traceability.

Step 5 — Versioning — The previous version is NEVER deleted; it is marked as superseded with a forward reference to the new version. The full history of constitutional amendments is retained for the regulatory retention period (7 years).

[L0] This process ensures full traceability — no constitutional change is ever silent. Every amendment is visible, reviewed, approved, and logged.

0.6 Constitutional Boundaries (What SGTX Is Not)

[L0] The following boundary statements clarify what SGTX is NOT. These are constitutional prohibitions — no component, configuration, or business decision may cross these boundaries.

[L0] SGTX is NOT a marketplace — There is no public listing of trades, no public ranking of providers, no public matching of buyers and sellers. Counterparty relationships are explicitly established and controlled by the trader.

[L0] SGTX is NOT a custodian — SGTX never holds customer funds. The FeeLock mechanism locks the fee amount but does not collect or hold the fee. All funds flow through connected banks and regulated PSPs.

[L0] SGTX is NOT a bank — SGTX does not accept deposits, issue credit, or execute settlement. It orchestrates payment instructions to banks; settlement confirmation is an Event from the bank, not from SGTX.

[L0] SGTX is NOT a customs authority — SGTX interfaces with customs authorities but never replaces them. Customs clearance is an authoritative act by the customs authority; SGTX only submits declarations on behalf of brokers.

[L0] SGTX is NOT a carrier — SGTX does not own transport assets (vehicles, vessels, aircraft). It orchestrates logistics execution through approved carriers, LSPs, and shipping lines.

[L0] SGTX is NOT a government — SGTX is not a government entity. It does not issue regulations, impose tariffs, or grant licences. It integrates with government systems that perform these functions.

[L0] SGTX is NOT an autonomous AI trader — AI assists but never forces. The AI subsystem cannot execute trades, release funds, or modify state without human authorization (A1-A3) or deterministic policy (A4). A5 (autonomous AI) is FORBIDDEN.

[L0] SGTX is NOT a title-taking intermediary — SGTX never takes title to goods. Title transfer is governed by the contract Incoterm and applicable law. SGTX records title transfer events but never becomes the title holder.

PART III — LAYER 1: ARCHITECTURE

[L1] This layer contains the current normative architecture of SGTX. These specifications describe how the platform is structured and how components interact. They are mutable through the standard change-management process (see Appendix C), but must always comply with Layer 0.

1. Multi-Clock State Vector Model

[L1] SGTX maintains a multi-clock state vector for every trade. Each domain has its own logical clock that advances independently, enabling the platform to reason about temporal consistency across heterogeneous processes that operate at different speeds and with different finality guarantees.

The state vector is the platform's answer to a fundamental problem of global trade: different processes (commercial, logistics, customs, financial) operate on different timelines and with different finality models. A single 'trade status' field cannot capture this complexity. Instead, the state vector tracks 12 independent domain clocks, each with its own finality class, and computes a divergence index that indicates how far apart the clocks have drifted.

The divergence index is a leading indicator of trade health. When all clocks are within 1 finality step of each other (NONE divergence), the trade is progressing normally. When clocks drift by 3+ steps (HIGH/CRITICAL divergence), the Governor intervenes — it may raise an exception, notify the relevant parties, or block further actions until the divergence is resolved.

1.1 The 12 Domain Clocks

#	Domain	Clock Tracks	Typical Finality Range	Governor Intervention
1	Commercial	Trade initiation, quote, contract, order, PO/SO, proforma	F0→F4 (initiation to contract signed)	Intervenes if stuck at F1 >7 days
2	Logistics	Transport booking, execution, tracking, delivery, milestone confirmation	F0→F5 (booking to delivery accepted)	Intervenes if shipment delayed >estimated transit +3 days
3	Customs	Declaration, assessment, clearance, post-clearance audit	F0→F4 (declaration to cleared)	Intervenes if held at F2 (submitted) >48 hours
4	Financial	Payment instructions, settlement legs, reconciliation	F0→F4 (instruction to settled)	Intervenes if settlement leg stuck at F2 >24 hours
5	Documentation	Document generation, verification, legalization, authentication	F0→F4 (draft to legalized)	Intervenes if mandatory doc missing at contract lock
6	Compliance	Sanctions screening, regulatory checks, KYB/AML	F0→F4 (screening to cleared)	Intervenes if sanctions POTENTIAL_MATCH — blocks all actions
7	Insurance	Policy quote, issuance, certificate, claims	F0→F4 (quote to policy issued)	Intervenes if insurance required but not issued before loading
8	Quality Control	Inspection scheduling, field inspection, reporting, certification	F0→F4 (scheduled to certified)	Intervenes if QC required but not completed before loading
9	Dispute	Filing, mediation, arbitration, resolution	F0→F4 (filed to resolved)	Intervenes if dispute open at closure — blocks USTN closure
10	Post-Trade	Returns, claims, warranty, drawback, post-clearance correction	F0→F4 (filed to resolved)	Intervenes if post-clearance open at closure — blocks unless severity ≤2

#	Domain	Clock Tracks	Typical Finality Range	Governor Intervention
11	Evidence	Evidence package assembly, hashing, sealing	F0→F5 (assembly to sealed)	Intervenes if evidence not sealed at closure — blocks USTN closure
12	Governance	Governor decisions, OPA evaluations, Loom entries, multisig approvals	F0→F5 (proposal to sealed)	Self-monitoring — Governor tracks its own decision latency

1.2 Finality Classes (F0–F5)

[L1] Each domain clock advances through 6 finality classes. A class represents the degree of certainty that a domain's state is final and will not change.

Class	Name	Meaning	Example (Commercial Domain)	Reversibility
F0	None	No activity — the domain has not started	Trade not yet initiated	N/A (nothing to reverse)
F1	Proposed	Intent has been expressed but no action taken	Buyer has expressed intent to trade (draft)	Easily reversible — withdraw intent
F2	Asserted	An action has been taken, awaiting confirmation	Trade request submitted to seller (PENDING_SELLER_RESPONSE)	Reversible with cost — withdraw request, notify seller
F3	Confirmed	Counterparty or authority has confirmed	Seller has submitted a quote (QUOTED)	Reversible with penalty — contract breach risk
F4	Settled	Financial settlement complete / authoritative confirmation received	Contract signed and locked (CONTRACT_SIGNED)	Very difficult to reverse — requires multisig + legal process
F5	Sealed	Immutable, evidence-sealed, post-closure observation active	USTN CLOSED — evidence package sealed	IRREVERSIBLE — recovery ≠ erasure; corrections via new events only

1.3 Divergence Index

[L1] The divergence index measures how far apart the 12 domain clocks have drifted. It is computed as the maximum finality-class difference between any two domains.

Index	Max Drift	Meaning	Governor Action	Trade Health
NONE	≤1 step	All domains progressing in sync	No action needed	HEALTHY
LOW	2 steps	Minor drift — one domain slightly ahead/behind	Log + monitor	HEALTHY (expected in normal trade)
MEDIUM	3 steps	Moderate drift — one domain significantly behind	Notify relevant parties	ATTENTION (may indicate a problem)
HIGH	4 steps	Significant drift — one domain severely behind	Raise exception (severity 2) + notify	WARNING (intervention likely needed)
CRITICAL	5+ steps	Severe drift — multiple domains stuck/blocked	Raise exception (severity 3+) + block further actions	CRITICAL (immediate intervention required)

[L1] The divergence index is recomputed every time any domain clock advances. If the index reaches CRITICAL, the Governor blocks all new actions on the trade until the divergence is resolved. This prevents trades from continuing to execute when fundamental processes are stuck.

[L1] **State integrity computation:** The platform computes a state integrity score (0-100) based on: divergence index (40% weight), number of open exceptions (30% weight), number of overdue Governor decisions (20% weight), and evidence completeness (10% weight). A score below 50 triggers a Governor review.

2. Event Spine (Immutable Event Log)

[L1] The Event Spine is the immutable, append-only, hash-chained log of every significant event in the platform. It is the single source of truth for what happened, when, and by whose authority. The Event Spine is never modified or deleted; corrections are made by appending new events that reference the original.

The Event Spine is the platform's memory. Every trade action — from initial intent through final closure — is recorded as an event. Events are hash-chained: each event's hash is computed over the previous event's hash plus the current event's payload. This creates a tamper-evident chain where any modification to a historical event breaks all subsequent hashes, making tampering immediately detectable.

2.1 Command ≠ Event Taxonomy

[L1] The platform enforces a strict separation between Commands and Events. This separation is fundamental to the architecture — it ensures that intent is always distinguished from fact, and that the audit trail records what actually happened, not what was attempted.

Aspect	Command	Event
Nature	Intent to change state	Fact that has occurred
Direction	Incoming (into Governor pipeline)	Outgoing (from Governor pipeline to Event Spine)
Tense	Imperative ('SubmitQuote','LockContract','CloseUSTN')	Past tense ('QuoteSubmitted','ContractLocked','USTNClosed')
Authority	Issued by a party (buyer, seller, system)	Emitted by the Governor after authorization
Reversibility	Can be withdrawn before processing	IRREVERSIBLE — once appended, never modified or deleted
Storage	Not stored (processed and discarded)	Stored permanently in Event Spine (7-year retention)
Hash chain	Not part of hash chain	Each event's hash includes the previous event's hash
Example API	POST /api/sgtx/quote/submit	GET /api/sgtx/events/[ustn] (retrieve event history)

2.2 Event Authority Taxonomy

[L1] Events are classified by their authority attribute — who may emit them and what level of trust they carry.

Authority Class	Who May Emit	Trust Level	Example Events
Observation	System (passive recording of external state)	Low — records what was observed, not what was confirmed	ContainerGateIn (terminal scanner), VesselDeparted (AIS feed)
Assertion	A party (buyer, seller, broker, carrier)	Medium — active claim by a known party, not yet confirmed	TradeInitiated (buyer), QuoteSubmitted (seller), DeclarationFiled (broker)

Authority Class	Who May Emit	Trust Level	Example Events
Confirmation	A trusted authority (customs, bank, government, Governor)	High — authoritative verification by a trusted source	CustomsCleared (customs authority), PaymentSettled (bank), ContractLocked (Governor)
System	Platform (automated system events)	Medium — system-generated based on rules or schedules	ExceptionRaised (system), RecoveryExecuted (Governor), SnapshotCaptured (system)

2.3 SHA-256 Hash Chain

[L1] Each event in the Event Spine is hash-chained to the previous event. The hash is computed as:

eventHash = SHA-256(previousEventHash + eventType + ustn + timestamp + actorGtid + canonicalPayloadJSON)

This creates a tamper-evident chain where:

- Any modification to a historical event changes its hash, which breaks the next event's hash, which breaks all subsequent hashes.
- The chain can be verified by recomputing all hashes from the genesis event and comparing to stored hashes.
- The chain is externally verifiable — any auditor with access to the Event Spine can verify integrity without trusting SGTX.
- The hash chain is replicated to 3 storage nodes across 2 jurisdictions (per the RTO/RPO targets in Part 15).

2.4 Replay Mode (§86)

[L1] The state vector for any USTN can be reconstructed by replaying all events from the Event Spine. This enables:

- **Audit:** Regulators and auditors can reconstruct the exact state of a trade at any point in time.
- **Dispute resolution:** If parties disagree on what happened, the Event Spine provides an immutable record.
- **Recovery:** If the state vector store is corrupted, it can be rebuilt by replaying events.
- **Testing:** The adversarial test suite uses event replay to verify state consistency.

Replay mode processes events in timestamp order, applying each event's effect to the state vector. The result is a fully reconstructed state vector that matches the live state (assuming no bugs in the replay logic).

3. Governor Pipeline

[L1] The Governor is the constitutional enforcement engine. Every irreversible action passes through the Governor pipeline, which evaluates the action against G1–G7 principles, OPA policies, and WasmEdge constitutional rules. The Governor is the gatekeeper — nothing bypasses it.

The Governor pipeline is designed to be deterministic, auditable, and fast. It evaluates each action through a series of gates, each of which returns a verdict (ALLOW, CONDITIONAL, or DENY). The final verdict is the strictest of all gates (DENY > CONDITIONAL > ALLOW). This ensures that a single gate's denial cannot be overridden by other gates' approvals.

3.1 Pipeline Stages

[L1] The Governor pipeline processes each Command through the following stages:

Stage	Name	Description	Verdict Impact
1	Command Received	A Command is submitted to the Governor via governorDecide().	—
2	G1 — Execution Gated	Verify that the action requires Governor approval (all irreversible actions do). If not irreversible, bypass remaining gates.	ALLOW if reversible
3	G2 — OPA Enforced	OPA evaluates the action against authored policies. Returns ALLOW/CONDITIONAL/DENY based on policy rules.	DENY if policy violation
4	G3 — WasmEdge Constitutional	WasmEdge executes constitutional rules (non-custody, non-marketplace, 110% reserve, closure-is-earned). Returns ALLOW/DENY.	DENY if constitutional violation
5	G5 — Multisig Check	If the action is irreversible (USTN closure, policy amendment, fund release), check multisig approval. 2-of-3 standard, 3-of-5 constitutional.	DENY if multisig not met
6	G6 — AI Advisory Check	If the AI subsystem has issued a recommendation (A1-A3), verify it has been reviewed. AI cannot block; it can only advise.	CONDITIONAL if AI flagged for review
7	Decision Merge	Merge all gate verdicts. Strictest wins: DENY > CONDITIONAL > ALLOW.	Final verdict
8a	If ALLOW	Execute the action + append Event to Event Spine + append decision to Loom.	—
8b	If CONDITIONAL	Execute the action with conditions recorded + append Event with conditions + append decision to Loom.	—
8c	If DENY	Block the action + append denial to Loom + return reason to caller.	—

3.2 Decision Merge Semantics

[L1] When multiple gates evaluate the same action, the strictest verdict wins. This is a critical safety property — it ensures that a single gate's denial cannot be overridden.

Gate A Verdict	Gate B Verdict	Merged Verdict	Rationale
ALLOW	ALLOW	ALLOW	All gates approve — action proceeds
ALLOW	CONDITIONAL	CONDITIONAL	At least one gate has conditions — conditions are recorded
ALLOW	DENY	DENY	One gate denies — action blocked
CONDITIONAL	CONDITIONAL	CONDITIONAL	Both gates have conditions — all conditions recorded
CONDITIONAL	DENY	DENY	One gate denies — action blocked

Gate A Verdict	Gate B Verdict	Merged Verdict	Rationale
DENY	DENY	DENY	All gates deny — action blocked

3.3 Governor Decision Lifecycle

[L1] Each Governor decision has a lifecycle:

- **Proposed:** A Command is submitted to the Governor.
- **Evaluating:** The Governor pipeline processes the Command through all gates.
- **Decided:** The Governor returns a verdict (ALLOW/CONDITIONAL/DENY) with conditions and rationale.
- **Executed:** If ALLOW/CONDITIONAL, the action is executed and an Event is appended to the Event Spine.
- **Logged:** The decision (including verdict, conditions, rationale, and all gate evaluations) is appended to the Loom.
- **Verifiable:** The Loom entry can be verified by any auditor — the hash chain proves the decision was not tampered with.

[L1] The Governor's decision latency target is <500ms for standard decisions and <5s for decisions requiring multisig (multisig latency depends on signatory availability).

4. Settlement Orchestration Control Plane

[L1] The Settlement Orchestration Control Plane manages multi-leg payment instructions. It orchestrates the flow of payment instructions to banks and PSPs, tracks settlement leg states, and enforces atomicity policies. The Control Plane is non-custodial — it sends instructions but never holds funds.

A single trade may have multiple payment legs: goods payment to the seller, freight payment to the carrier, duty payment to customs, SGTX fee payment to the platform, QC inspection fee to the QC provider, lab test fee to the laboratory. Each leg is an independent payment instruction with its own state machine, its own bank integration, and its own settlement confirmation.

4.1 Multi-Leg Payment Model

[L1] The multi-leg model allows a trade to have N independent payment legs, each with its own:

- **Beneficiary:** The recipient (seller, carrier, customs authority, SGTX, QC provider, lab)
- **Amount:** The payment amount in the specified currency
- **Bank/PSP:** The financial institution processing the payment
- **State machine:** PENDING → SUBMITTED → ACCEPTED → SETTLED (or REJECTED / RETURNED / REVERSED)
- **Settlement confirmation:** An Event from the bank confirming settlement (not from SGTX)

State	Meaning	Trigger	Reversibility
PENDING	Payment instruction created but not yet sent to bank	Leg created by Settlement Orchestration	Easily reversible (cancel instruction)
SUBMITTED	Payment instruction sent to bank	Bank Settlement Gateway submits to bank API	Reversible with cost (recall request to bank)
ACCEPTED	Bank has accepted the instruction for processing	Bank returns acceptance (API response or webhook)	Reversible with penalty (cancellation fee)
SETTLED	Bank confirms settlement complete — funds transferred	Bank returns settlement confirmation (Event)	IRREVERSIBLE (funds have moved)
REJECTED	Bank rejected the instruction	Bank returns rejection (compliance, AML, insufficient funds)	Reversible (fix issue + resubmit)
RETURNED	Payment was returned by the beneficiary bank	Bank returns return notification	Reversible (investigate + resubmit)
REVERSED	Payment was reversed after settlement (chargeback)	Bank returns reversal notification	IRREVERSIBLE (reversal is final)

4.2 Atomicity Policies

[L1] The Settlement Orchestration Control Plane supports 5 atomicity policies that determine how multiple legs interact:

Policy	Meaning	Use Case	Failure Handling
ALL_OR_NONE	All legs must settle, or all fail. If any leg fails, all successful legs are reversed.	Goods payment + SGTX fee (fee should only settle if goods payment settles)	If any leg REJECTED/RETURNED, reverse all SETTLED legs
PARTIAL_ALLOWED	Legs may settle independently. No requirement for all to settle.	Independent service fees (QC fee, lab fee — separate from goods payment)	No rollback — each leg is independent

Policy	Meaning	Use Case	Failure Handling
SEQUENCED	Legs must settle in a specified order. Leg N+1 cannot settle until Leg N settles.	Goods payment → Freight payment (freight paid after goods confirmed)	Block Leg N+1 until Leg N SETTLED
CONDITIONAL	Legs settle based on conditions (e.g., delivery accepted, customs cleared).	Duty payment (conditional on customs clearance) → SGTX fee (conditional on duty payment)	Block until condition met
HUMAN_RELEASE	Legs require human release before settlement.	Large payments above threshold, disputed payments	Block until human releases via multisig

4.3 Bank Settlement Gateway (6-Stage Pipeline)

[L1] The Bank Settlement Gateway processes each payment instruction through a 6-stage pipeline before submitting to the bank. This pipeline is simulated — real bank APIs are called by the connected bank's integration, not by SGTX directly.

Stage	Name	Description	Failure Handling
1	Schema Validation	Validate the payment instruction against ISO 20022 or bank-specific format.	REJECT if schema invalid — return error to caller
2	Signature Validation	Verify the QES (Qualified Electronic Signature) or bank API authentication.	REJECT if signature invalid — return auth error
3	USTN Validation	Verify the trade exists, is not blocked, and the payment leg is valid for this USTN.	REJECT if USTN invalid or blocked — return validation error
4	Beneficiary Consistency	Verify the beneficiary matches the trade contract (seller, carrier, customs, etc.).	REJECT if beneficiary mismatch — return consistency error
5	Bank Policy Check	Check bank-specific policies (transaction limits, currency restrictions, cut-off times).	REJECT if policy violation — return policy error
6	AML / Sanctions Screening	Screen the payment against AML rules and sanctions lists (UN, OFAC, EU).	REJECT if sanctions match — return compliance error + raise exception

4.4 Non-Custody Boundary

[L0-1, L1] **SGTX orchestrates settlement instructions but NEVER holds funds.** This is the architectural non-custody boundary, enforced by WasmEdge constitutional modules.

The FeeLock mechanism works as follows:

- **At trade initiation:** The SGTX fee (1.5% of trade value) is calculated and 'locked' — the amount is recorded but NOT collected.
- **At contract lock:** The fee amount is included in the settlement instruction sent to the bank.
- **At settlement:** The bank settles the goods payment to the seller AND the SGTX fee to SGTX's bank account — simultaneously. SGTX receives the fee only after the bank confirms settlement.
- **At no point:** Does SGTX hold the trade value, the seller's payment, or any customer funds. The bank is the custodian throughout.

[L1] This architecture makes non-custody a structural property, not a policy choice. The code path for SGTX to receive funds simply does not exist — funds flow from buyer's bank to seller's bank (and SGTX's bank for the fee) via the bank's settlement system.

5. 28-Add-On Catalogue (Full Details)

[L1] The SGTX platform includes 28 add-on modules. Each add-on extends the core platform with specific capabilities. This section documents each add-on in full: purpose, database schema, API endpoints, integration points, and implementation checklist.

The status matrix uses the v14.0 deployment-state vocabulary: CORE_READY (implemented and tested), PRODUCTION_CONNECTED (live integration active), LEGAL_AUTHORIZATION_REQUIRED (technical ready, awaiting legal/regulatory approval).

Add-On 1: GNN Risk Engine & Institutional Trade Graph

Priority: Foundation | **Status:** CORE_READY | **Blueprint Ref:** Part 11.1

Purpose: Graph Neural Network for sanctions-proximity detection (UBO 2-hop) and trust-based trade-graph mapping. The GNN analyses the network of known counterparties to identify sanctions proximity and compute trust scores.

Database Schema: GnnModel, TrustGraphNode, TrustGraphEdge, SanctionsProximityScore

API Endpoints: POST /api/sgtx/gnn/train, GET /api/sgtx/gnn/trust-score/{gtid}, GET /api/sgtx/gnn/sanctions-proximity/{gtid}

Integration: Part 4.3 (Product Form Agent — trust display), Part 12A.2 (TCC — trust card), Part 10 (Dispute — trust evidence)

> **Notes:** [L0-25] Non-marketplace bounded: may score trust for known parties, NEVER discovers/recommends/ranks counterparties.

Add-On 2: Federated Learning Network

Priority: Foundation | **Status:** CORE_READY | **Blueprint Ref:** Part 11.2

Purpose: Train fraud/margin/credit models across sovereign nodes without raw data sharing. Each sovereign node trains local models on its own memory events; only encrypted model updates are shared.

Database Schema: FederatedModel, FederatedNode, ModelUpdate, TrainingRound

API Endpoints: POST /api/sgtx/federated/train, GET /api/sgtx/federated/models, POST /api/sgtx/federated/contribute

Integration: Part 19 (Trade Memory Layer — training data source), Part 12A.2 (TCC — model predictions)

> **Notes:** Differential privacy (epsilon=0.5) enforced. No raw trade data leaves the sovereign node.

Add-On 3: Causal Inference Engine

Priority: Foundation | **Status:** CORE_READY | **Blueprint Ref:** Part 11.3

Purpose: DoWhy + EconML root-cause attribution for disputes, milestone breaches, and quality failures. Identifies the causal factors behind trade exceptions (e.g., '68% of temperature excursions caused by carrier delay, not equipment failure').

Database Schema: CausalAnalysis, CausalFactor, InterventionResult

API Endpoints: POST /api/sgtx/causal/analyse, GET /api/sgtx/causal/factors/{ustn}, GET /api/sgtx/causal/interventions

Integration: Part 10 (Dispute — root cause evidence), Part 12A.2 (TCC — causal analysis card)

> **Notes:** A2 authority — proposes causal factors; humans decide whether to act on them.

Add-On 4: Self-Healing Infrastructure & Chaos Engineering

Priority: Foundation **| Status:** CORE_READY **| Blueprint Ref:** Part 11.4

Purpose: Automated chaos testing + self-healing. The platform injects controlled failures (network partition, DB timeout, API outage) and verifies that recovery mechanisms work. Self-healing automatically restarts failed components.

Database Schema: ChaosTest, ChaosResult, SelfHealingAction, RecoveryLog

API Endpoints: POST /api/sgtx/chaos/run, GET /api/sgtx/chaos/results, GET /api/sgtx/chaos/health

Integration: Part 12C.11 (Admin Portal — chaos test management), Part 18 (Observability)

> **Notes:** Chaos tests run in sandbox environment only — never in production without explicit approval.

Add-On 5: Automated Penetration Testing

Priority: Foundation **| Status:** CORE_READY **| Blueprint Ref:** Part 11.5

Purpose: Continuous automated penetration testing. The platform runs scheduled pentests against its own APIs, infrastructure, and integrations. Findings are tracked as ThreatFindings with severity + remediation SLA.

Database Schema: PentestRun, ThreatFinding, RemediationAction

API Endpoints: POST /api/sgtx/pentest/run, GET /api/sgtx/pentest/findings, POST /api/sgtx/pentest/remediate

Integration: Part 12C.11 (Admin Portal — threat findings), Part 18 (Observability)

> **Notes:** A3 authority — escalates findings to human security team; never auto-remediates severity 4+.

Add-On 6: Post-Quantum Cryptography (PQC)

Priority: Foundation **| Status:** CORE_READY **| Blueprint Ref:** Part 11.6

Purpose: Dilithium3 for archival records (quantum-resistant signatures). PQC is used for long-term evidence storage (7-year retention) to protect against future quantum attacks on classical cryptography.

Database Schema: PqcKey, PqcSignature, ArchivalRecord

API Endpoints: POST /api/sgtx/pqc/sign, GET /api/sgtx/pqc/verify/{id}, POST /api/sgtx/pqc/rotate-keys

Integration: Part 22 (Post-closure evidence sealing), Part 18 (Security architecture)

> **Notes:** PQC is additive — classical signatures remain for short-term verification; PQC for archival.

Add-On 7: Expanded ZK Proofs & Proof of Reserves

Priority: Foundation | **Status:** CORE_READY | **Blueprint Ref:** Part 11.7

Purpose: Zero-knowledge proofs for trust graph queries (client-side WASM via Plonky3) and proof-of-reserves attestation. ZK proofs allow SGTX to prove properties (reserve ratio ≥ 110%, trust score > threshold) without revealing the underlying data.

Database Schema: ZkProof, ReserveAttestation, ReserveComposition

API Endpoints: POST /api/sgtx/zk/generate, GET /api/sgtx/zk/verify/{id}, GET /api/sgtx/zk/reserves

Integration: Part 4.3 (Product Form Agent — trust query), Appendix A (Financial — reserve attestation)

> **Notes:** [L0-17, L0-23] Reserve tables store attestations only — not custody. ZK proves 110% without revealing amounts.

Add-On 8: Customs Bond & Guarantee Management

Priority: P0 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.8

Purpose: Manage customs bonds, bank guarantees, and comprehensive guarantees for transit. Tracks bond sufficiency, renewal dates, and claim history. Jurisdiction-aware (Egypt, EU, US, UAE, Saudi, UK).

Database Schema: CustomsBond, BondSufficiencyCheck, BondClaim, JurisdictionBondRule

API Endpoints: POST /api/sgtx/bonds/create, GET /api/sgtx/bonds/sufficiency/{ustn}, POST /api/sgtx/bonds/claim

Integration: Part 5.7 (Customs Declaration — bond requirement), Part 12C.10 (Gov Portal — bond oversight)

> **Notes:** Bond factor varies by jurisdiction (EG 1.0x, EU 0.3-1.0x, US 1.0x, AE 0.5x).

Add-On 9: Demurrage & Detention Management

Priority: P0 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.9

Purpose: Auto-extracts carrier tariffs, tracks free time real-time per port/container/carrier, proactive 48h/24h/expiry alerts, jurisdiction-aware norms. A2 AI predicts congestion; A4 enforces settlement.

Database Schema: DemurrageTracking, CarrierDemurrageTariff, PortFreeTime, DemurrageAlert

API Endpoints: GET /api/sgtx/demurrage/{ustn}, POST /api/sgtx/demurrage/calculate, GET /api/sgtx/demurrage/alerts

Integration: Part 3B.3.5.3 (Mode C carrier quotes — tariff extraction), Part 6 (Payment — demurrage in settlement), Part 8 (Container Release), Part 12A.1 (Smart Inbox — alerts)

> **Notes:** Proactive alerts: 48h (priority 95), 24h (95), expiry (90), escalated (85).

Add-On 10: Broker Liability & Insurance Management

Priority: P0 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.10

Purpose: Under Egyptian Customs Law 207/2020, brokers are personally liable for declaration accuracy. This module tracks broker liability insurance, declaration errors, and performance metrics. Mandatory insurance min EGP 500,000 (Egypt).

Database Schema: BrokerLiabilityInsurance, BrokerDeclarationError, BrokerPerformanceMetric

API Endpoints: POST /api/sgtx/broker-liability/create, GET /api/sgtx/broker-liability/status, GET /api/sgtx/broker-liability/performance

Integration: Part 9.5 (CBR Portal — insurance upload + error tracking), Part 2.2 (Onboarding), Part 5.7 (Customs Declaration — error logging)

> **Notes:** Penalties: 5-50% of duty value for declaration errors. Insurance must be renewed annually.

Add-On 11: Customs Valuation Intelligence

Priority: P1 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.11

Purpose: AI-driven duty estimation (A2 XGBoost) and market-value comparison. Real-time comparison declared vs. market average; proactive alerts when deviation >20%; valuation history stored for audit defence.

Database Schema: CustomsValuation, ValuationDispute, MarketPriceData

API Endpoints: GET /api/sgtx/valuation/{ustn}, POST /api/sgtx/valuation/calculate, POST /api/sgtx/valuation/dispute, GET /api/sgtx/valuation/market-price

Integration: Part 4.15 (Governor PreScreen — valuation gate G1U35), Part 4.3 (Product Form Agent — market data), Part 10 (Dispute — valuation category)

> **Notes:** Deviation >20% triggers ALERT; >50% triggers ENHANCED_DD. Market data from RIA (GRiRE).

Add-On 12: Cold Chain Quality Management

Priority: P1 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.12

Purpose: EU RASFF-compliant continuous temperature monitoring (15-min intervals, 5-year retention), PTI certificate tracking, data logger calibration, A2 LSTM anomaly detection.

Database Schema: ColdChainRequirement, PtiCertificate, ColdChainReading, ColdChainAnomaly, DataLoggerCalibration

API Endpoints: GET /api/sgtx/cold-chain/requirements, POST /api/sgtx/cold-chain/pti/register, GET /api/sgtx/cold-chain/status/{ustn}, POST /api/sgtx/cold-chain/reading, GET /api/sgtx/cold-chain/anomalies/{ustn}

Integration: Part 4.3 (Product Form Agent), Part 4.5 (Documentation), Part 13.1.25 (IoT Sensor — enhanced anomaly detection)

> **Notes:** Temperature readings every 15 minutes. Anomaly severity: LOW (<2°C, <30min), MEDIUM (<2°C, >30min), HIGH (>2°C or >1h).

Add-On 13: Inspection Agency Accreditation

Priority: P1 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.13

Purpose: Track ISO 17020 accreditation, scope of work, performance metrics for QC inspection agencies. A1 performance summaries; A2 trend analysis; A4 enforcement (block jobs for unaccredited agencies).

Database Schema: InspectionAgencyAccreditation, InspectionAgencyPerformance, AccreditationBody

API Endpoints: GET /api/sgtx/inspection/accreditations, POST /api/sgtx/inspection/accredit, GET /api/sgtx/inspection/performance/{agencyGtid}

Integration: Part 9.4 (QC Portal — accreditation check before job assignment), Part 2.2 (Onboarding), Part 12C.10 (Gov Portal — accreditation view)

> **Notes:** Accreditation must be valid (not expired). Expired accreditation blocks new job assignment.

Add-On 14: Currency Risk Management

Priority: P1 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.14

Purpose: Real-time FX rates from ECB (free XML daily feed), hedging recommendations, natural hedging opportunity matching. Advisory only — never forces hedging. A2 volatility prediction.

Database Schema: CurrencyExposure, HedgingRecommendation, FxRate

API Endpoints: GET /api/sgtx/currency-risk/exposure/{ustn}, GET /api/sgtx/currency-risk/recommendations, GET /api/sgtx/currency-risk/history

Integration: Part 4.9 (Commercial Settlement — currency selection), Part 5.6 (Invoice — multi-currency), Part 6 (Payment — FX in settlements), Part 12A.2 (TCC — currency risk card)

> **Notes:** ECB rates updated daily at 16:00 CET. Hedging recommendations advisory (A1) — trader decides.

Add-On 15: Government API Sandbox

Priority: P1 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.15

Purpose: Government APIs change without notice. This module provides a sandbox environment mirroring production, automated daily regression tests against live APIs, change detection alerts, and adaptive integration auto-update for minor changes.

Database Schema: GovernmentApiSandbox, GovernmentApiTestResult, ApiChangeDetectionLog

API Endpoints: GET /api/sgtx/gov-sandbox/apis, POST /api/sgtx/gov-sandbox/test, GET /api/sgtx/gov-sandbox/results, POST /api/sgtx/gov-sandbox/sync

Integration: Part 7 (Government Integration — sandbox for testing), Part 12C.10 (Gov Portal — sandbox status), Part 12C.11 (Admin Portal — sandbox management)

> **Notes:** 10 APIs with sandbox: Nafeza (EG), CargoX (EG), ETA (EG), ICS2 (EU), TRACES (EU), Trade Single Window (UAE), FASAH (Saudi), ACE (USA), CDS (UK).

Add-On 16: FTA Preference Management

Priority: P1 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.16

Purpose: Auto-detect FTA eligibility, calculate preference rates, manage certificates (EUR.1, COO, etc.). A2 FTA detection; A4 validation.

Database Schema: FtaPreference, FtaPreferenceClaim, FtaPreferenceRule

API Endpoints: GET /api/sgtx/fta/preferences, GET /api/sgtx/fta/check, POST /api/sgtx/fta/claim, GET /api/sgtx/fta/certificate

Integration: Part 4.3 (Product Form Agent), Part 4.5 (Documentation), Part 5.7 (Customs Declaration — FTA claim), Part 12A.2 (TCC — FTA savings display)

> **Notes:** 9 FTAs supported: Egypt-EU (0% EUR.1), Egypt-UAE, Egypt-Saudi, Egypt-UK, AfCFTA, GAFTA, EU-Mercosur, USMCA, RCEP.

Add-On 17: Piracy & Security Risk Engine

Priority: P1 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.17

Purpose: Maritime security intelligence, corridor scoring, insurance premium adjustment for high-risk routes. Sources: IMB Piracy Reporting Centre, Maritime Security Council. Advisory only — never blocks trades. 4 risk levels with insurance impact.

Database Schema: MaritimeSecurityIncident, CorridorSecurityScore, SecurityAdvisory

API Endpoints: GET /api/sgtx/security/corridor/{code}, GET /api/sgtx/security/incidents, GET /api/sgtx/security/advisories, GET /api/sgtx/security/insurance-impact

Integration: Part 30 (Trade Corridors — security scoring), Part 4.8 (Insurance — premium adjustment), Part 12A.2 (TCC — security risk card)

> **Notes:** Risk levels: LOW (0%), MODERATE (+10%), HIGH (+30%), CRITICAL (+100%) insurance premium impact.

Add-On 18: Trade Compliance Calendar

Priority: P1 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.18

Purpose: Centralised calendar for regulatory deadlines, certificate expiries, tariff changes. Proactive reminders 30/14/7/1 days before. Auto-population by RIA (regulatory change detection).

Database Schema: ComplianceCalendarEvent, ComplianceCalendarAlert, RegulatoryDeadline

API Endpoints: GET /api/sgtx/compliance-calendar/events, GET /api/sgtx/compliance-calendar/upcoming, POST /api/sgtx/compliance-calendar/event, POST /api/sgtx/compliance-calendar/complete

Integration: Part 4.1 (RIA — deadline detection), Part 12A.1 (Smart Inbox — reminders), Part 12A.10 (Task Center), Part 12C.10 (Gov Portal — compliance dashboard)

> **Notes:** Event types: Tariff Change, Certificate Expiry, Licence Renewal, Regulatory Reporting, Sanctions Update, Trade Agreement Change, Bond Expiry. Reminders: 30, 14, 7, 1 days before.

Add-On 19: Cargo Insurance Integration

Priority: P2 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.19

Purpose: API integration with insurance providers — premium calculation, policy issuance, claim handling. A2 premium optimisation; A4 validation.

Database Schema: InsuranceProvider, InsurancePolicy, InsuranceClaim

API Endpoints: GET /api/sgtx/cargo-insurance/premium, POST /api/sgtx/cargo-insurance/policy/issue, GET /api/sgtx/cargo-insurance/policy/{ustn}, POST /api/sgtx/cargo-insurance/claim/submit

Integration: Part 4.8 (Insurance Requirements), Part 5.6 (Invoice — premium in invoice), Part 12A.2 (TCC — insurance card), Part 10 (Dispute — claim evidence)

> **Notes:** Providers: Allianz, Zurich, AIG (global, API ■); Egyptian Insurance (Egypt); Saudi National (Saudi).

Add-On 20: Trade Finance Documentation

Priority: P2 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.20

Purpose: Manage LC applications, confirmations, bills of exchange, assignment of proceeds. Workflow tracking with status/approvals/signatures. Integrates with financing module.

Database Schema: TradeFinanceDocument, TradeFinanceCase

API Endpoints: POST /api/sgtx/trade-finance/document, GET /api/sgtx/trade-finance/documents/{ustn}, POST /api/sgtx/trade-finance/document/sign

Integration: Part 4.9 (Commercial Settlement — LC selection), Part 3B.5 (Financing Module — LC confirmation), Part 6 (Payment Orchestrator — bill of exchange)

> **Notes:** Document types: LC Application, LC Confirmation, Bill of Exchange, Assignment of Proceeds, Standby LC.

Add-On 21: Back-to-Back LC Management

Priority: P2 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.21

Purpose: Manage chains of LCs (primary → secondary). Buyer uses credit line to issue LC to seller; seller uses it to issue second LC to supplier. Chain tracking; risk management; coordinated execution.

Database Schema: BackToBackLc, LetterOfCredit, LcLifecycle

API Endpoints: POST /api/sgtx/back-to-back-lc/create, GET /api/sgtx/back-to-back-lc/{id}, GET /api/sgtx/back-to-back-lc/chain/{ustn}

Integration: Part 20 (Trade Finance Docs — LC management), Part 3B.5 (Financing Module — financing chain), Part 12A.2 (TCC — LC chain card)

> **Notes:** Primary LC + Secondary LC linked. Risk: if primary LC is cancelled, secondary LC must be cancelled too.

Add-On 22: Force Majeure Handling

Priority: P2 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.22

Purpose: Handle force majeure events (natural disasters, war, pandemic) affecting trade execution. Event detection via RIA scraping official sources (WHO for pandemic); structured claim submission; contract renegotiation; jurisdiction-aware clauses.

Database Schema: ForceMajeureEvent, ForceMajeureClaim, ForceMajeureExtension

API Endpoints: GET /api/sgtx/force-majeure/events, POST /api/sgtx/force-majeure/claim, GET /api/sgtx/force-majeure/claim/{id}, POST /api/sgtx/force-majeure/extension

Integration: Part 4.1 (RIA — event detection), Part 3 (Contract — extension clauses), Part 10 (Dispute — force majeure evidence), Part 12A.1 (Smart Inbox — event alerts)

> **Notes:** Event types: Natural Disaster, War/Conflict, Pandemic (WHO alert), Port Strike, Sanctions. Claims require evidence (official notice, news report).

Add-On 23: Shipper's Declaration & Export Documentation

Priority: P2 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.23

Purpose: Manage shipper's declarations, export accompanying documents (EAD), and export licence tracking. A2 document extraction; A4 validation.

Database Schema: ShippersDeclaration, ExportAccompanyingDocument, ExportLicence

API Endpoints: POST /api/sgtx/shippers-declaration/create, GET /api/sgtx/shippers-declaration/{ustn}, POST /api/sgtx/shippers-declaration/ead, GET /api/sgtx/shippers-declaration/licence/status

Integration: Part 5.7 (Customs Declaration — declaration data), Part 2.2 (Onboarding — licence upload), Part 12A.1 (Smart Inbox — licence expiry)

> **Notes:** Documents: Shipper's Declaration (all exports), EAD (EU exports), Export Licence (restricted goods), COO (FTA preference), EUR.1 (Egypt-EU FTA).

Add-On 24: Port & Terminal Integration

Priority: P2 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.24

Purpose: Beyond container release API — deeper integration with port/terminal systems. EDI integration (gate-in/gate-out); pre-advice; terminal OS real-time status. A4 validation only.

Database Schema: TerminalIntegration, GateEvent, TerminalEvent

API Endpoints: POST /api/sgtx/terminal/pre-advice, POST /api/sgtx/terminal/gate-in, POST /api/sgtx/terminal/gate-out, GET /api/sgtx/terminal/status/{container}

Integration: Part 8 (Container Release — gate events trigger release), Part 3B.5.1 (Pre-advice — webhook), Part 13.1.5 (Ports — terminal mapping)

> **Notes:** Integration points: Gate-in (EDIFACT), Gate-out (EDIFACT), Vessel Schedule (API/EDI), Container Status (API/EDI), Pre-advice (API).

Add-On 25: Payment Guarantee Confirmation (Optional)

Priority: P3 | **Status:** CORE_READY | **Blueprint Ref:** Part 11.25

Purpose: OPTIONAL — payment guarantee verification is NOT mandatory. Traders can choose not to use it. Provides optional verification of payment guarantees (LCs, bank guarantees) through SWIFT MT700/URDG 758. Multiple methods: SWIFT, manual, third-party.

Database Schema: PaymentGuaranteeConfirmation

API Endpoints: POST /api/sgtx/payment-guarantee/create, POST /api/sgtx/payment-guarantee/confirm, GET /api/sgtx/payment-guarantee/status

Integration: Part 12A.2 (TCC — optional card), Part 20 (Trade Finance Docs — guarantee confirmation)

> **Notes:** Verification methods: SWIFT MT700 (LC), URDG 758 (bank guarantee), Manual Upload (PDF), Third-Party (independent verification).

Add-On 26: Demurrage Dispute Resolution

Priority: P3 **| Status:** CORE_READY **| Blueprint Ref:** Part 11.26

Purpose: Demurrage disputes are common — this module provides structured dispute resolution workflow integrated with Part 10 main dispute system. Structured workflow (initiate, evidence, mediate, resolve); automatic demurrage calculation audit trail as evidence package.

Database Schema: DemurrageDispute

API Endpoints: POST /api/sgtx/demurrage-dispute/create, GET /api/sgtx/demurrage-dispute/list

Integration: Part 10 (Dispute Management — dispute category), Part 9 (Demurrage — evidence package), Part 12A.2 (TCC — dispute card)

> **Notes:** Dispute reasons: FREE_TIME_MISSED, RATE_MISMATCH, WRONG_CONTAINER_TYPE, CARRIER_ERROR, PORT_CONGESTION, FORCE_MAJURE, DOCUMENTATION_ERROR, DOUBLE_CHARGE, OTHER.

Add-On 27: (RESERVED)

Priority: — | Status: — | Blueprint Ref: —

Purpose: This add-on is reserved for future enhancements. No schema, no endpoints, no checklist. Placeholder for future expansion.

Database Schema: —

API Endpoints: —

Integration: —

> **Notes:** Correctly reserved per blueprint — no stubs or half-baked implementations exist.

Add-On 28: GRiRE (Global Regulatory Intelligence & Requirements Engine)

Priority: Foundation **| Status:** CORE_READY **| Blueprint Ref:** Part 11.28

Purpose: AI-powered discovery and import of regulatory, customs, documentation, and logistics requirements for every country/territory worldwide. 4-layer pipeline: Scraper (Rust+Rig) → AI Parsing (A2 HF NLP) → Structured Store (PostgreSQL+Vector) → Output (dynamic forms, checklists, calculators).

Database Schema: CountryRegulatoryProfile, HsTariffRate, CountryRequiredDocument, ColdChainRequirement, FtaPreferenceRule, RegulatoryChangeLog, GrireSource

API Endpoints: GET /api/sgtx/grire/country-profile, GET /api/sgtx/grire/tariff, GET /api/sgtx/grire/required-docs, GET /api/sgtx/grire/cold-chain, GET /api/sgtx/grire/fta-preference, GET /api/sgtx/grire/full-report, POST /api/sgtx/grire/discover

Integration: Part 4.1 (RIA — GRiRE is the advanced AI-powered version), Part 4.3 (Product Form Agent), Part 4.5 (Documentation), Part 7 (Government Integration — API discovery), Part 13 (Data Model — jurisdiction-specific tables)

Notes: Coverage: 195 countries, 150+ tariff schedules, 180+ document matrices, 160+ cold chain profiles, 200+ port demurrage norms, 350+ FTA rules. New country onboarded in <1 day with GRiRE.

6. Jurisdiction Capability Adapter Schema

[L1] SGTX supports a worldwide jurisdiction capability adapter fabric. Each jurisdiction (country, customs territory, economic union) has a profile that defines its regulatory requirements, customs procedures, payment rails, and government integrations.

The jurisdiction model goes beyond simple 'country' — it supports 16 jurisdiction types: sovereign country, customs territory, customs union, economic union, autonomous territory, special administrative region, free zone, free port, bonded zone, special economic zone, airport customs jurisdiction, port customs jurisdiction, tax territory, export-control territory, special regime, and more.

6.1 16 Connector States

[L1] Every government integration progresses through 16 states, from initial discovery through production connection to deprecation:

State	Description
NOT_DISCOVERED	No knowledge of the government system exists.
DISCOVERED	System identified but not yet documented.
DOCUMENTED	System documented (API specs, EDI formats, portal URLs).
CONTACT_REQUIRED	Contact with the government authority needed.
CREDENTIALS_REQUIRED	API credentials / certificates needed.
SANDBOX_AVAILABLE	Sandbox environment available for testing.
SANDBOX_CONNECTED	Sandbox integration active and tested.
CERTIFICATION_REQUIRED	Government certification needed for production.
CERTIFICATION_PENDING	Certification application submitted, awaiting approval.
PRODUCTION_READY	Technical, legal, operational, commercial all ready.
PRODUCTION_CONNECTED	Live production integration active.
DEGRADED	Integration partially functional (some endpoints failing).
OUTAGE	Integration completely non-functional.
PORTAL_ONLY	No API — process via government web portal manually.
MANUAL_ONLY	No electronic integration — process via physical visit / paper.
DEPRECATED	Integration retired — no longer supported.

6.2 7 Authoritative Statuses

[L1] Separately from connector states, the government's authoritative status for a trade action is tracked. SGTX never invents government approval — it only records what the government has actually said.

Status	Meaning
SGTX_READY	SGTX has prepared the submission but not yet sent it to the government.
SUBMITTED	SGTX has submitted to the government system.
GOVERNMENT_ACCEPTED	Government has accepted the submission (processing).
GOVERNMENT_REJECTED	Government has rejected the submission (correction needed).
GOVERNMENT_HOLD	Government has placed the submission on hold (review/inspection).

Status	Meaning
GOVERNMENT_RELEASED	Government has released/cleared the submission.
MANUAL_AUTHORITY_CONFIRMED	A human has manually confirmed the authority's decision (fallback).

6.3 4-Dimension External Readiness

[L1] Every integration reports readiness across 4 independent dimensions. 'Connected' requires ALL FOUR dimensions to be satisfied. This prevents the false claim 'connected' when only the technical dimension is met.

Dimension	Question	Evidence Required
TECHNICAL	Is the API/EDI integration working?	API response logs, successful test transactions, uptime metrics
LEGAL	Are contracts/agreements signed?	Signed legal agreement, data processing agreement, NDA
OPERATIONAL	Are procedures tested and documented?	Runbook, trained operators, tested failure scenarios
COMMERCIAL	Are fees and commercial terms agreed?	Signed commercial agreement, fee schedule, SLA

[L1] **Example:** If the TECHNICAL dimension is met (API works) but the LEGAL dimension is not (contract not signed), the integration status is 'SANDBOX_CONNECTED' — not 'PRODUCTION_CONNECTED'. The 4-dimension model ensures honest reporting.

7. Regulatory Classification Gate

[L1] The Regulatory Classification Gate maps the platform's actual functionality to the applicable licence class in each jurisdiction. This is a dynamic, jurisdiction-aware classification — not a static label.

[L0-22] **Non-custody is architectural, not legal classification.** SGTX's non-custody property is an architectural fact (the platform never holds customer funds). However, actual functionality determines licensing in each jurisdiction. A jurisdiction may classify SGTX's orchestration role as requiring specific licences regardless of the non-custody architecture.

[L1] The classification matrix starts with Egypt (CBE Law 194/2020, Nafeza customs, ETA tax) and is extensible to other jurisdictions via the Jurisdiction Capability Adapter. The matrix maps each SGTX functionality to: (1) the applicable Egyptian law/regulation, (2) the required licence/registration, (3) the responsible authority, and (4) the current compliance status.

8. Closure Policy

[L1] A USTN is closed only when all 7 closure conditions are met. Closure is earned, not forced. The canClose predicate is a pure function that evaluates all 7 conditions and returns true only if all are satisfied.

#	Condition	Description	Evidence Required
1	Delivery Accepted	Final delivery confirmed by receiver — quantity, condition, quality verified.	Proof of Delivery (POD) signed by receiver, including condition assessment.
2	Settlement Complete	All payment legs settled (goods, freight, duty, SGTX fee, service fees).	Bank settlement confirmation Events for all legs.
3	Financial Reconciliation Complete	Bank/PSP reconciliation matches SGTX records — no material mismatch.	Reconciliation report showing all legs match within tolerance (<$1).
4	Customs Complete	Import and export customs cleared, no open holds.	Customs clearance Events from customs authority.
5	Post-Clearance Complete	Post-clearance audit, correction, refund/drawback processed if applicable.	Post-clearance audit closure or 'no audit required' confirmation.
6	Disputes/Claims Satisfied	All filed disputes and claims resolved or time-barred.	Dispute resolution Events for all open disputes.
7	Evidence Sealed	26-category evidence package assembled, hashed, and sealed (immutable).	Evidence package hash + sealing Event from Governor.

[L1] **canClose predicate:** canClose(ustn) = AND(conditions[1..7] all met). The Governor evaluates this predicate; it never auto-closes. If canClose returns true, a human (multisig 2-of-3) triggers the closure ceremony.

[L1] **CLOSED_WITH_EXCEPTION:** In limited cases, a USTN may be closed with an open exception if the sole blocker is an EXCEPTION_OPEN state with severity ≤ 2 (low). Severity 3–5 exceptions block closure. The severity matrix ensures that minor administrative exceptions do not indefinitely block closure, while material exceptions do.

[L1] **Post-closure event rules (§22):** After closure, a post-closure observation period is active (default 90 days, extendable). New evidence may be added (e.g., late-arriving customs confirmation) but sealed evidence cannot be modified. The Transaction Twin continues to observe and may trigger re-opening if a material exception (severity 3+) is discovered during the observation period.

9. AI Recommendation Gateway

[L1] The AI Recommendation Gateway channels AI outputs through the Governor pipeline. AI never executes directly; it proposes, and the Governor decides. This is the constitutional boundary [L0-9, L0-21].

[L1] **Flow:** (1) AI subsystem (A1–A3) produces a recommendation. (2) Recommendation is submitted to the Governor as a Command. (3) Governor evaluates against G1–G7, OPA policies, WasmEdge rules. (4) If ALLOW, the recommended action is executed by the deterministic policy engine (A4). (5) If CONDITIONAL, conditions are recorded. (6) If DENY, the recommendation is blocked + reason logged. (7) The event is appended to the Loom.

[L1] **Multi-model consensus:** The AI Brain uses 3 independent providers (Gemini, Groq, HuggingFace) for consensus-based recommendations. Each provider is queried in parallel; results are merged via weighted voting (Gemini 0.4, Groq 0.35, HuggingFace 0.25). A minimum of 2 providers must agree for A2/A3 recommendations. This prevents single-provider bias and improves recommendation quality.

10. Transport Engine Architecture

[L1] SGTX supports 5 transport modes as first-class engines, each with its own entity types, state machines, and terminal adapters. A Multimodal Orchestrator coordinates multi-leg journeys across modes. The USTN remains the canonical reference across all modes.

10.1 Road Corridor Engine (Articles 43-46)

[L1] The Road Corridor Engine handles international trucking across multiple countries. It supports Egypt, Jordan, Saudi, UAE, GCC, Iraq, Libya, and future jurisdictions. Key entities:

Entity	Purpose	Key Fields
RoadCorridor	Top-level corridor (e.g., Egypt → Jordan → Saudi → UAE)	corridorCode, originCountry, destinationCountry, transitCountries, totalDistanceKm, estimatedTransitHours
RoadLeg	A single leg within a corridor (e.g., Cairo → Aqaba border)	corridorId, sequence, originLocation, destinationLocation, borderCrossing, distanceKm, estimatedHours
RoadShipment	A specific shipment moving through a corridor	ustn, corridorId, carrierGtid, vehicleId, driverId, status (PLANNED→IN_TRANSIT→AT_BORDER→CLEARED→DELIVERED)
RoadVehicle	Registered truck/trailer	vehicleRegistration, vehicleType (TRUCK/TRAILER/TRACTOR), capacityKg, insuranceValidUntil, dgCapability, reeferCapability
RoadDriver	Authorized driver	fullName, passportNumber, licenseNumber, licenseValidUntil, visaCountries, dgAuthorization
RoadBorderCrossing	Border crossing event	shipmentId, borderName, country, crossingType (EXIT/ENTRY/TRANSIT), customsDeclarationRef, sealNumber
RoadGpsTracking	GPS ping for a shipment	shipmentId, latitude, longitude, recordedAt, speed, heading

10.2 Air Cargo Engine (Articles 47-52)

[L1] The Air Cargo Engine handles air freight with IATA ONE Record compatibility. Key entities:

Entity	Purpose	Key Fields
AirBooking	Top-level air booking	ustn, bookingReference, shipperGtid, consigneeGtid, originAirport, destinationAirport, flightDate, mawbNumber
AirFlight	Flight record	flightNumber, airline, originAirport, destinationAirport, scheduledDeparture, scheduledArrival, aircraftType
AirAirport	Airport reference	iataCode (e.g., CAI), icaoCode, name, city, country, timezone
AirWaybill	MAWB / HAWB	bookingId, waybillType (MAWB/HAWB), waybillNumber, shipper, consignee
AirPiece	Individual cargo piece	bookingId, pieceNumber, sscc, weightKg, lengthCm, widthCm, heightCm, volumeCbm
AirUld	Unit Load Device (container)	bookingId, uldNumber, uldType (AKE/PAJ/PMC), tareWeightKg, maxPayloadKg, contents
AirStatusEvent	Milestone status event (RCS/DEP/ARR/RCF/NFD/DLV)	bookingId, eventType, eventTime, airport, remarks
AirChargeableWeight	Chargeable weight calculation	bookingId, actualWeightKg, volumetricWeightKg, chargeableWeightKg, ratePerKg, totalCharge

[L1] **Air status event types:** RCS (Received for Shipment), DEP (Departed), ARR (Arrived), RCF (Received at Consignee Facility), NFD (Notified for Delivery), DLV (Delivered). These are the IATA standard milestone codes.

[L1] **Chargeable weight:** The greater of actual weight and volumetric weight. Volumetric weight = (length × width × height in cm) / 6000 (IATA standard divisor).

10.3 Ocean Container Engine (Article 53)

[L1] The Ocean Container Engine handles containerised sea freight. Key entities: Booking, Vessel, Voyage, Port, Container, VGM (Verified Gross Mass), B/L, e-B/L, Manifest, ACI, Terminal, Gate, Transshipment, Demurrage, Detention, Delivery. The engine integrates with Add-On 9 (Demurrage) for cost tracking and Add-On 24 (Port & Terminal) for gate events.

10.4 RoRo & Rolling Cargo Engine (Articles 55-86)

[L1] The RoRo Engine is the largest transport engine, covering 32 blueprint articles. It handles vehicles and rolling cargo with VIN-level tracking. Key entities:

Entity	Purpose	Key Fields
RoRoShipment	Top-level master object	ustn, shipmentReference, shipperGtid, consigneeGtid, originPort, destinationPort, totalUnits, totalWeightKg
RoRoUnit	Individual rolling cargo unit (VIN-level)	shipmentId, vin, unitType (VEHICLE/TRUCK/TRACTOR/TRAILER/BUS/MOTORCYCLE/MACHINERY), make, model, year, weightKg, runningStatus
RoRoVoyage	Vessel voyage	vesselName, vesselImo, voyageNumber, operatorGtid, originPort, destinationPort, etd, eta, bookingCutoff, gateCutoff
RoRoBooking	Booking on a voyage	shipmentId, voyageId, bookingReference, unitsCount, totalWeightKg, preferredSailing, deliveryWindow
RoRoYard	Yard position tracking	unitId, yardZone, block, row, slot, deck, position, status (EXPECTED→ARRIVED→GATE_IN→INSPECTED→PARKED→READY_FOR_LOADING→LOADED→DISCHARGED→AVAILABLE→RELEASED→GATE_OUT)

Entity	Purpose	Key Fields
RoRoGateEvent	Gate-in / gate-out event	shipmentId, unitId, eventType (GATE_IN/GATE_OUT), gateType (ORIGIN/DESTINATION), eventTime, vinScan, customsStatus
RoRoInspection	Pre-load / post-discharge inspection	unitId, inspectionType (PRE_LOAD/POST_DISCHARGE/CLAIM), inspectorName, mileage, fuelLevel, preExistingDamage, newDamage
RoRoBillOfLading	B/L for the shipment	shipmentId, blNumber, blType (MASTER/HOUSE), shipper, consignee, vesselName, voyageNumber, vinsList

[L1] **19-state unit state machine:** BOOKED → DOCUMENTS_PENDING → CUSTOMS_PENDING → READY_FOR_GATE → GATE_IN → INSPECTION_PENDING → INSPECTED → YARD → READY_FOR_LOAD → LOADED → AT_SEA → TRANSSHIPMENT → DISCHARGED → DESTINATION_YARD → CUSTOMS_HOLD → CUSTOMS_RELEASED → DELIVERY_ORDER → READY_FOR_GATE_OUT → GATE_OUT → DELIVERED → ACCEPTED.

[L1] **12-state vessel state machine:** SCHEDULED → BOOKING_OPEN → CUTOFF_APPROACHING → CARGO_ACCEPTING → LOADING → DEPARTED → AT_SEA → TRANSSHIPMENT → ARRIVED → DISCHARGING → COMPLETED. Never mix with unit state.

[L1] **Damage comparison:** AI (A2) marks POSSIBLE_DAMAGE; human marks CONFIRMED_DAMAGE. States: NO_CHANGE, POSSIBLE_DAMAGE, CONFIRMED_DAMAGE, DISPUTED_DAMAGE. AI never determines liability autonomously.

10.5 Rail Engine (Article 54)

[L1] The Rail Engine handles rail freight with CIM/SMGS consignment note support. Key entities: RailBooking, RailTrain, RailWagon, RailTerminal, RailConsignment, RailTransit, RailStatusEvent.

[L1] **Consignment note types:** CIM (EU rail) and SMGS (former Soviet rail). The engine supports both formats and can convert between them for cross-border rail movements.

10.6 Multimodal Orchestrator

[L1] The Multimodal Orchestrator coordinates multi-leg journeys across transport modes. A single USTN may span: Truck → RoRo → Rail → Truck (e.g., Egypt → Damietta → Trieste → Rail → Rotterdam → Truck → Final Destination). The orchestrator manages the handoff between modes, ensuring that the USTN remains the canonical reference across all legs.

[L1] **Transport hierarchy:** ROAD, AIR, OCEAN_CONTAINER, RORO, RAIL, FERRY, MULTIMODAL — all beneath one transport orchestrator and one USTN graph. Each specialist engine owns mode-specific rules; the orchestrator coordinates cross-mode handoffs.

11. Canonical Data Model

[L1] The SGTX Canonical Data Model consists of the following primary stores. Each store has a specific purpose and a defined schema. The stores are designed to be independently queryable while maintaining cross-references via the USTN.

Store	Purpose	Key Fields	Retention
State Vector Store	12-domain × F0-F5 finality tracking per USTN	ustn, domainClocks[12], finalityClass, divergenceIndex, healthScore	7 years
Event Spine	Immutable hash-chained event log	eventId, previousEventHash, eventType, ustn, timestamp, actorGtid, payload, signature	7 years
Obligation Graph	Directed dependency graph of trade obligations	obligationId, ustn, type, state, dependencies[], transitiveImpact	7 years
Settlement Instructions/Legs	Multi-leg payment instruction tracking	instructionId, ustn, legs[], atomicityPolicy, state	7 years
External Identifier Registry	17 identifier types (UCR, MAWB, HAWB, B/L, etc.)	identifierId, type, value, ustn, lifecycle, issuingAuthority	7 years
Recovery Vault	Content-addressable (SHA-256) evidence storage	entryId, ustn, type, contentHash, reference, verified	7 years
Transaction Twin	14-domain digital twin for post-closure observation	twinId, ustn, domains[14], postClosureActive, observationExpiry	7 years
Financial Exposure	14-dimension exposure tracking per USTN	exposureId, ustn, dimensions[14], state, outstandingAmount	7 years
Exception Events	Severity 1-5 exception tracking with SLA	exceptionId, ustn, category, severity, slaDeadline, resolutionAction	7 years
Closure Policy	7-condition closure evaluation per USTN	policyId, ustn, conditions[7], blockers[], canClose, closureState	7 years
Trade Stage Log	36-stage lifecycle tracking per USTN	id, ustn, stageCode, stageName, completedAt, completedBy	7 years
Quote	Dedicated quote entity (replaces JSON-in-Trade.globalNotes)	id, ustn, quoteNumber, totalQuote, exwPrice, sgtxFee, lineItems, status	7 years

12. Provider Relationship Model

[L1] The provider relationship model governs how traders connect to service providers (LSPs, carriers, brokers, QC, labs, insurers, financiers). The model is relationship-controlled, not marketplace.

[L1] **Relationship types:** A provider can become available to a trader because: (1) it is already approved and connected (platform-level approval), (2) the trader has saved its GTID (trader-level save), (3) the trader explicitly selected it (per-trade selection), or (4) a government-mandated service relationship exists (e.g., mandatory port authority).

[L1] **Prohibited:** random provider matching, unsolicited provider recommendations, 'best provider' suggestions, public provider rankings, public provider marketplace, autonomous provider selection. These are all blocked by the non-marketplace principle [L0-2, L0-16, L0-25].

[L1] **Workflow:** Trader → Approved/Connected/Saved Provider → RFQ/Quote → Trader Review → Trader Explicit Selection → Service Agreement/Addendum → Execution. The trader always makes the final selection — the platform never auto-selects a provider.

13. Master Global Trade Graph

[L1] The entire SGTX platform is represented as a directed graph: GTID → TRADE → RFQ → QUOTATION → ORDER → CONTRACT → REGULATORY_SNAPSHOT → USTN → TRANSPORT_GRAPH → CUSTOMS_OPERATIONS → DOCUMENTS → GOVERNMENT_REFERENCES → PAYMENT/FINANCE → PHYSICAL_EXECUTION → DELIVERY → ACCEPTANCE → SETTLEMENT → RECONCILIATION → POST-CLEARANCE → CLAIMS/RETURNS → EVIDENCE → USTN_CLOSED.

[L1] Every node in this graph references the USTN. The graph is traversable in both directions — from a USTN, you can find all related nodes; from any node, you can find the parent USTN. This enables comprehensive audit trails and cross-referencing.

PART IV — LAYER 2: IMPLEMENTATION SPECIFICATIONS

[L2] This layer contains concrete, testable implementation artefacts. These specifications are designed to be directly implementable by an engineering team without ambiguity.

14. Canonical Event Type Catalogue

[L2] The following event types are canonical. Each event has required fields, an authority class (who may emit it), a clock impact (which domain clock it advances), and lineage rules (what it references).

Event Type	Authority	Clock Impact	Required Fields	Lineage
TradeInitiated	Assertion (Buyer)	Commercial F1	ustn, buyerGtid, sellerGtid, commodity, incoterm, tradeValueUsd	References: buyerGtid, sellerGtid
QuoteSubmitted	Assertion (Seller)	Commercial F2	ustn, quoteId, totalQuote, exwPrice, sgtxFee, lineItems	References: ustn, sellerGtid
QuoteAccepted	Assertion (Buyer)	Commercial F3	ustn, quoteId, acceptedAt	References: ustn, quoteId
NegotiationStarted	Assertion (Party)	Commercial F2	ustn, round, proposalType, proposalDetails	References: ustn, quoteId
PurchaseOrderCreated	Assertion (Buyer)	Commercial F3	ustn, poNumber, totalValue, lineItems	References: ustn, quoteId
SalesOrderCreated	Assertion (Seller)	Commercial F3	ustn, soNumber, poId, totalValue	References: ustn, poId
ProformaIssued	Assertion (Seller)	Commercial F3	ustn, proformaNumber, totalAmount, validUntil	References: ustn, soId
ContractLocked	Confirmation (Governor)	Commercial F4, Governance F3	ustn, multisigId, conditions[]	References: ustn, all prior
RegulatorySnapshotCaptured	Observation (System)	Compliance F3	ustn, snapshotHash, tariffRate, sanctionsStatus	References: ustn, jurisdiction
ShipmentBooked	Assertion (Shipper)	Logistics F2	ustn, bookingRef, mode, origin, destination	References: ustn, contractId
ContainerLoaded	Observation (Terminal)	Logistics F3	ustn, containerNumber, vesselName, voyageNumber	References: ustn, bookingRef
VesselDeparted	Observation (Carrier)	Logistics F3	ustn, vesselName, voyageNumber, actualDeparture	References: ustn, bookingRef
CustomsDeclarationSubmitted	Assertion (Broker)	Customs F2	ustn, declarationRef, declarationType	References: ustn, shipmentId
CustomsCleared	Confirmation (Authority)	Customs F4	ustn, clearanceRef, clearedAt	References: ustn, declarationRef
PaymentInstructionSubmitted	Assertion (Payer)	Financial F2	ustn, legId, amount, currency, beneficiary	References: ustn, bankRef
PaymentSettled	Confirmation (Bank)	Financial F4	ustn, legId, bankRef, settledAt	References: ustn, legId
InspectionCompleted	Assertion (QC/Lab)	QC F3	ustn, inspectionId, result, inspector	References: ustn, shipmentId

Event Type	Authority	Clock Impact	Required Fields	Lineage
DeliveryAccepted	Assertion (Receiver)	Logistics F5	ustn, podRef, acceptedAt, condition	References: ustn, shipmentId
DisputeFiled	Assertion (Party)	Dispute F2	ustn, disputeId, reason, filedBy	References: ustn, evidence[]
ExceptionRaised	Observation (System)	Governance F2	ustn, exceptionId, category, severity	References: ustn, trigger
RecoveryExecuted	Confirmation (Governor)	Governance F3	ustn, exceptionId, recoveryPath, recoveredAt	References: ustn, exceptionId
EvidenceSealed	Confirmation (Governor)	Evidence F5	ustn, evidenceHash, sealedAt, categories[26]	References: ustn, all
USTNClosed	Confirmation (Governor)	All domains F5	ustn, closedAt, closedBy, conditionsMet[7]	References: ustn, all
TradeStageCompleted	Observation (System)	Governance F3	ustn, stageCode, stageName, completedAt, completedBy	References: ustn, prior stage

[L2] **Required fields for ALL events:** eventId, eventType, ustn, timestamp, actorGtid, authorityClass, previousEventHash, payload (JSON), signature (QES or system). Events without these fields are rejected by the Event Spine.

15. API Contract Structure

[L2] The SGTX API enforces Command/Event separation. Commands are POST/PUT requests that intent to change state. Events are GET requests that retrieve immutable records. No API endpoint both reads and writes state.

[L2] **Standard response envelope:** All API responses use { ok: boolean, data: ..., error?: string, filter?: object }. List endpoints return { ok, , count, filter }. This is the v14.0 standardized contract — all transport engines (Road, Air, RoRo, Rail) return this shape.

[L2] **Idempotency:** All POST endpoints accept an Idempotency-Key header. If the same key is sent twice, the second request returns the first request's response without re-executing. This prevents duplicate trade creation on network retry.

[L2] **Rate limiting:** API endpoints are rate-limited per tenant GTID. Standard limit: 100 requests/minute. AI-intensive endpoints: 20 requests/minute. Rate limit headers (X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Reset) are included in every response.

16. Observability Catalogue

[L2] SGTX maintains the following observability surfaces:

Surface	Metrics	Retention	Alerting
Platform Health	Request rate, error rate, latency p50/p95/p99, uptime	30 days	SEV-0 if uptime <99.5%
Trade Metrics	Trades initiated, USTN closure rate, avg trade value, avg time-to-close	7 years	SEV-1 if closure rate <80%
Governor Metrics	Decisions/minute, DENY rate, CONDITIONAL rate, multisig latency	7 years	SEV-1 if DENY rate >10%
Event Spine Metrics	Events/minute, hash chain verification, replay success rate	7 years	SEV-0 if hash chain breaks
Settlement Metrics	Settlement leg states, bank API latency, reconciliation match rate	7 years	SEV-1 if reconciliation mismatch >$1
Exception Metrics	Open exceptions by severity, SLA breach count, recovery success rate	7 years	SEV-1 if any severity 4+ exception open >24h
AI Metrics	Recommendations/minute, acceptance rate, fallback rate, provider latency	30 days	SEV-2 if provider latency >15s
Integration Metrics	Connector health (16 states), API success rate, sandbox vs production	90 days	SEV-1 if production connector DEGRADED

[L2] **Minimum failure-path test suite:** The platform maintains an adversarial test suite covering:

#	Test	Expected Result
1	USTN closure with missing conditions	Must block — canClose returns false
2	Non-custody violation attempt	Must block — WasmEdge detects fund-holding code path
3	AI A5 autonomy attempt	Must block — AI cannot call state-mutating APIs directly
4	Marketplace matching attempt	Must block — no public listing/ranking/recommendation endpoints
5	Event spine tampering	Must detect — hash chain verification fails

#	Test	Expected Result
6	Governor bypass attempt	Must block — all state mutations call governorDecide()
7	Settlement without bank confirmation	Must block — settlement requires bank Event, not SGTX confirmation
8	Recovery that erases history	Must block — Loom is append-only, corrections via new events
9	Closure with open severity 3+ exception	Must block — CLOSED_WITH_EXCEPTION only for severity ≤2
10	Reserve ratio below 110%	Must block — WasmEdge constitutional rule

17. RTO/RPO Targets

[L2] Recovery Time Objective and Recovery Point Objective targets:

Tier	RTO	RPO	Scope	Durability
Critical	4 hours	0 (append-only)	Event Spine, State Vector, Governor decisions	3 copies, 2 jurisdictions, quorum 2-of-3
Standard	24 hours	0 (append-only)	All trade data, settlement instructions, evidence	3 copies, 2 jurisdictions, quorum 2-of-3
Extended	72 hours	< 1 hour	Add-on data, analytics, logs	2 copies, 1 jurisdiction, single-write

[L2] **Durability definition:** '3 copies, 2 jurisdictions, quorum 2-of-3' means data is replicated to 3 storage nodes across at least 2 legal jurisdictions, and writes require acknowledgement from 2 of 3 nodes before acknowledgement. This ensures no single jurisdiction's legal action can make data unavailable.

18. Security Architecture

[L2] Security is enforced at multiple layers:

Layer	Mechanism	Enforcement
Network	mTLS between all services, network isolation, DDoS protection	Infrastructure-level — all inter-service traffic is mTLS
Authentication	QES (Qualified Electronic Signature) for all authoritative acts	WasmEdge verifies QES before executing irreversible actions
Authorization	OPA policies per endpoint + multisig for irreversible	Governor pipeline evaluates OPA + multisig before execution
Data at rest	AES-256 encryption for all databases	Database-level — Turso/libsql encryption
Data in transit	TLS 1.3 for all API calls	Infrastructure-level — no HTTP, only HTTPS
Audit	Loom hash chain + 7-year retention	Immutable — append-only, externally verifiable
PQC	Dilithium3 for archival records (Add-On 6)	Long-term protection against quantum attacks
Secrets	HSM (Hardware Security Module) for key storage	Infrastructure-level — no keys in code or config files

19. Global Standards Gateway

[L2] SGTX integrates with 20 global standards for interoperability:

Standard	Purpose	Direction
WCO Data Model	Customs data interoperability	SGTX → WCO mapping (output)
WCO Code Lists	Standardised code lists (HS, country, currency)	SGTX uses WCO codes (input)
HS (Harmonized System)	Commodity classification	SGTX uses HS6+ (input/output)
UN/CEFACT	Trade facilitation standards	SGTX → UN/CEFACT mapping
UN/LOCODE	Port/airport location codes	SGTX uses UN/LOCODE (input)
UN/EDIFACT	EDI message format	SGTX → EDIFACT (output for legacy systems)

Standard	Purpose	Direction
UBL	Universal Business Language	SGTX → UBL (output for electronic documents)
e-CMR	Electronic Consignment Note (road)	SGTX → e-CMR (output)
e-AWB	Electronic Air Waybill	SGTX → e-AWB (output)
e-B/L	Electronic Bill of Lading	SGTX → e-B/L (output)
Cargo-XML	IATA cargo XML standard	SGTX → Cargo-XML (output)
ONE Record	IATA single digital shipment view	SGTX ↔ ONE Record (bidirectional)
ISO 20022	Financial messaging standard	SGTX → ISO 20022 (output for bank integration)
XAdES/CAdES/PAdES	Advanced electronic signatures	SGTX uses for QES
GS1/EPCIS	Supply chain event tracking	SGTX → EPCIS (output)
Verifiable Credentials	Decentralised identity	SGTX → VC (output for tenant credentials)

PART V — PORTAL DOCUMENTATION (12 Portals)

[L2] This section documents all 12 SGTX portals in full. Each portal's tabs, screens, workflows, and data access patterns are documented. This is the comprehensive user-facing surface documentation that enables an engineering team to implement each portal without ambiguity.

[L2] The SGTX platform has 178 total tab entries across 12 portals. Each tab is a distinct screen with a specific purpose. Every tab is wired to a dispatcher branch in PortalContent.tsx — 100% wiring coverage.

20.1 Buyer Portal — Full Workflow

Portal ID: trader-buyer | **Tenant:** European Importer GmbH (SGTX-DE-TRD-001234-5B6C) | **Role:** Importer | **Tagline:** Import · Inbound · Settlement

1-Click Actions: 1-Click New Trade (→ new-trade tab) | 1-Click Pay Invoice (→ invoices tab)

Tabs (33 total):

Group	Tab	Purpose	Screen
Overview	Command Center	Executive summary + 1-Click Action Bar + Trade Lifecycle Visualizer + Smart Inbox	CommandCenter
Trade	New Trade Request	11-step wizard: Parties → Commodity → Containers → Documentation → Transport → Insurance → Settlement → Criticality → Shipments → Compliance Gates → Governor & Submit	NewTradeRequestScreen
Trade	Active Trades	List of trades in active execution (PENDING_SELLER_RESPONSE through IN_EXECUTION)	BuyerActiveTradesScreen
Trade	Drafts	Trades awaiting seller response (editable/cancellable)	BuyerDraftsScreen
Trade	History	Closed, settled, cancelled, rejected trades (read-only audit)	BuyerHistoryScreen
Trade	Quote Review & Negotiation	Review seller quotes, accept/reject/counter	QuoteReviewScreen
Trade	Negotiations	Active and historical negotiations (round, proposal type, status)	NegotiationsScreen
Trade	Purchase Orders	Buyer-created POs (PO number, seller, value, status)	PurchaseOrdersScreen
Trade	Proforma Invoices	Proforma invoices received from sellers	ProformaInvoicesScreen
Trade	Contract Signing	Sign contract addenda, view contract status	ContractSigningScreen
Trade	Shipments	Track inbound shipments (vessel, ETA, milestones)	ShipmentsVault
Trade	Container Compliance	Per-container compliance status (VGM, seal, manifest)	ContainerComplianceScreen
Trade	Milestone Tracking	Track delivery milestones (departure, transit, arrival)	MilestoneScreen
Trade	Reefer Monitoring	Real-time reefer temperature + humidity + anomalies	ReeferTelemetryPanel
Trade	Documents	Trade document repository (invoice, B/L, COO, certificates)	DocumentsScreen
Trade	Distressed Cargo	Distressed cargo listings + liquidation options	DistressedScreen

Group	Tab	Purpose	Screen
Trade	Routes Reference	Worldwide port routes reference (carrier, transit time, frequency)	RoutesReferenceScreen
Trade	Demurrage	Demurrage tracking + alerts (Add-On 9)	DemurragePanel
Trade	Cold Chain	Cold chain compliance + PTI + readings (Add-On 12)	ColdChainPanel
Finance	Financing (Borrower)	Request trade financing from connected banks/PFIs	FinancingBorrowerScreen
Finance	Invoices & Payments	View invoices, approve payment, track settlement	InvoicesScreen
Finance	FX & Settlement	Multi-currency settlement + FX rates + reconciliation	SettlementScreen
Governance	Disputes	File disputes, track dispute resolution	DisputesScreen
Governance	Compliance	Sanctions screening, regulatory checks, KYB status	ComplianceScreen
Governance	Audit Trail	Immutable event history for the tenant's trades	AuditScreen
Governance	Network (Contacts)	Saved counterparty network (no public marketplace)	NetworkScreen
Governance	Trust Passport	Tenant's trust portrait + KYB tier + trust score	PassportScreen
Governance	Trade Readiness	Readiness score + missing items (advisory, non-blocking)	ReadinessScreen
Governance	Tenant Lifecycle	Onboarding status + KYB verification + lifecycle state	LifecycleScreen
Governance	Compliance Calendar	Regulatory deadlines + certificate expiries (Add-On 18)	ComplianceCalendarPanel
Admin	Org Graph	Organisational graph (employees, roles, permissions)	OrgGraphScreen
Admin	GTID Chat	Cross-tenant secure messaging	ChatScreen
Admin	Company Admin	Company settings + employee management + bank details	CompanyAdminScreen

Buyer workflow (end-to-end):

- **1. Initiate trade:** Click '1-Click New Trade' → 11-step wizard → submit → trade created with PENDING_SELLER_RESPONSE status + seller notified + government notified.
- **2. Review quote:** Seller submits quote → buyer sees it in 'Quote Review' → accept/reject/counter. If counter, negotiation round created.
- **3. Create PO:** After quote acceptance → create Purchase Order → seller creates matching Sales Order.
- **4. Sign contract:** Review contract addenda → sign → Governor locks contract → USTN minted → regulatory snapshot captured.
- **5. Track shipment:** Monitor shipment progress via 'Shipments' + 'Milestone Tracking' + 'Reefer Monitoring' (if cold chain).
- **6. Approve payment:** Click '1-Click Pay Invoice' → review invoice → approve → payment instruction sent to bank → settlement tracked.
- **7. Accept delivery:** Receive goods → verify quantity/condition → accept delivery → POD signed.
- **8. Close USTN:** Click '1-Click Close USTN' → 7-condition check → if all met, USTN closed + evidence sealed.

20.2 Seller Portal — Full Workflow

Portal ID: trader-seller | **Tenant:** Strawberry Export Co. (SGTX-EG-TRD-002139-7F3A) | **Role:** Exporter | **Tagline:** Export · Outbound · Pricing

1-Click Actions: 1-Click Submit Quote (→ quote-builder tab) | 1-Click Request Payout (→ invoices tab)

Tabs (31 total):

Group	Tab	Purpose	Screen
Overview	Command Center	Executive summary + Seller Control Tower + 1-Click Action Bar	CommandCenter
Trade	Pending Requests	Inbound trade requests from buyers	SellerPendingRequestsScreen
Trade	Quote & Packing	Build EXW quote + packing plan + commodity line items	QuoteBuilderScreen
Trade	Negotiations	Active negotiations with buyers	NegotiationsScreen
Trade	Sales Orders	Sales orders created from accepted POs	SalesOrdersScreen
Trade	Proforma Invoices	Proforma invoices issued to buyers	ProformaInvoicesScreen
Trade	Contract & Addenda	Sign contract + manage addenda	ContractSigningScreen
Trade	Outbound Shipments	Track outbound shipments (loading, departure, transit)	ShipmentsVault
Trade	Container Compliance	Per-container compliance (VGM, seal, packing list)	ContainerComplianceScreen
Trade	Milestone Tracking	Confirm outbound milestones (loaded, departed, arrived)	MilestoneScreen
Trade	Documents	Trade document generation + upload	DocumentsScreen
Trade	Distressed Cargo	List distressed cargo for liquidation	DistressedScreen
Trade	Routes Reference	Worldwide port routes reference	RoutesReferenceScreen
Trade	Lot Management	Lot-level inventory management (for export grading)	LotManagementPanel
Trade	Demurrage	Demurrage tracking for outbound containers	DemurragePanel
Trade	Cold Chain	Cold chain compliance for perishable exports	ColdChainPanel
Finance	Financing (Borrower)	Request export financing (pre-shipment, post-shipment)	FinancingBorrowerScreen
Finance	Invoices & SGTX Fee	Issue invoices + track SGTX fee (1.5%)	InvoicesScreen
Finance	FX & Settlement	Multi-currency settlement + FX hedging (Add-On 14)	SettlementScreen
Governance	Disputes	Respond to disputes + provide evidence	DisputesScreen
Governance	Compliance & KYB	KYB status + compliance screenings	ComplianceScreen
Governance	Audit Trail	Immutable event history	AuditScreen
Governance	Network (Contacts)	Saved counterparty network	NetworkScreen
Governance	Trust Passport	Tenant trust portrait + export credentials	PassportScreen
Governance	Trade Readiness	Readiness score for export operations	ReadinessScreen
Governance	Tenant Lifecycle	Onboarding + export licence tracking	LifecycleScreen
Admin	Org Graph	Organisational graph	OrgGraphScreen
Admin	GTID Chat	Cross-tenant messaging	ChatScreen
Admin	Company Admin	Company settings + export licence management	CompanyAdminScreen

Seller workflow (end-to-end):

- **1. Receive request:** Buyer submits trade request → seller sees it in 'Pending Requests' + Smart Inbox notification.
- **2. Build quote:** Click '1-Click Submit Quote' → quote builder → set EXW price + packing plan + commodity lines → submit → buyer notified + government notified.
- **3. Negotiate:** If buyer counters → negotiation round created → respond with counter-offer or accept.
- **4. Create SO:** After PO received from buyer → create matching Sales Order.
- **5. Issue proforma:** Issue Proforma Invoice to buyer → buyer accepts → contract preparation begins.
- **6. Sign contract:** Review + sign contract addenda → Governor locks contract → USTN minted.
- **7. Prepare shipment:** Pack goods → load container → confirm VGM + seal → submit loading confirmation.
- **8. Track outbound:** Monitor shipment departure + transit + arrival at destination.
- **9. Receive payment:** Bank settles payment → seller notified → invoice marked paid.

20.3 LSP Portal — Full Workflow

Portal ID: lsp | **Tenant:** Delta Freight & Forwarding (SGTX-EG-LSP-000120-4C7D) | **Role:** Trucking & Forwarding | **Tagline:** Export · Pickup · Trucking · Milestones

1-Click Actions: 1-Click Dispatch (→ dispatch-planner) | 1-Click Log Milestone (→ milestones)

Tabs (15 total):

Group	Tab	Purpose
Overview	Command Center	Executive summary + active jobs + fleet status
Operations	Assignments	Active truck/driver assignments
Operations	Milestone Confirmation	Confirm pickup/departure/arrival/delivery milestones
Operations	Addenda	Manage transport addenda (CMR, waybills)
Operations	Fleet	Vehicle fleet management + maintenance
Operations	Dispatch Planner	Plan + dispatch trucks to pickup/delivery jobs
Operations	Warehouse	Warehouse dashboard + inventory
Operations	Performance	LSP performance metrics (on-time %, SLA)
Operations	Road Corridor	Road corridor management (Add-On: Road Engine)
Operations	Rail	Rail booking management (Add-On: Rail Engine)
Trade	Worldwide Routes	Worldwide port route search + sync
Finance	Invoices	Invoice management + payment tracking
Governance	Compliance	Vehicle/driver compliance + insurance
Governance	Audit Trail	Event history
Admin	Company Admin	Company settings

Workflow: Receive assignment → Dispatch truck → Confirm pickup → Track transit → Confirm border crossing → Confirm delivery → Log milestone → Issue invoice

20.4 Shipping Line Portal — Full Workflow

Portal ID: ship | **Tenant:** Maersk Levant (SGTX-EG-SHP-000031-9E8F) | **Role:** Ocean Carrier | **Tagline:** Vessels · Containers · B/L · Schedules

1-Click Actions: 1-Click Issue B/L (→ bl) | 1-Click Authorize Release (→ containers)

Tabs (14 total):

Group	Tab	Purpose
Overview	Command Center	Executive summary + vessel schedule + container status
Operations	Vessels	Vessel fleet management
Operations	Containers	Container inventory + release management
Operations	B/L (Bill of Lading)	Issue + manage B/L and e-B/L
Operations	Schedules & AIS	Vessel schedules + AIS tracking
Operations	Booking Requests	Inbound booking requests from shippers
Operations	Contract Rates	Manage contracted freight rates
Operations	Air Cargo	Air cargo booking management (Air Engine)
Operations	RoRo Cargo	RoRo shipment + unit + voyage management (RoRo Engine)
Operations	Performance	Carrier performance metrics
Trade	Worldwide Routes	Worldwide port route management
Finance	Invoices	Freight invoice management
Governance	Compliance	Vessel compliance + flag state
Governance	Audit Trail	Event history

Workflow: Receive booking → Assign vessel/voyage → Issue B/L → Track vessel (AIS) → Confirm arrival → Authorize container release → Issue freight invoice

20.5 QC Portal — Full Workflow

Portal ID: qc | **Tenant:** Nile Quality (SGTX-EG-QC-000022-8A1C) | **Role:** Pre-shipment Inspection | **Tagline:** Inspection · Quality · Certification

1-Click Actions: 1-Click Start Inspection (→ schedule) | 1-Click Issue Report (→ reports)

Tabs (10 total):

Group	Tab	Purpose
Overview	Command Center	Executive summary + inspection queue
Operations	Inspection Schedule	Schedule inspections + assign inspectors
Operations	Field Inspections	Conduct field inspections + log defects + upload photos
Operations	QC Reports	Issue QC reports + certificates
Operations	Re-Inspections	Manage re-inspection requests
Operations	Performance	Inspector performance metrics
Trade	Container Compliance	Per-container QC compliance
Finance	Invoices	QC fee invoicing
Governance	Compliance	Inspector accreditation (Add-On 13)
Governance	Audit Trail	Event history

Workflow: Receive inspection request → Schedule inspection → Assign inspector → Conduct field inspection → Log defects + photos → Issue QC report → Notify parties

20.6 Customs Broker Portal — Full Workflow

Portal ID: cbr | **Tenant:** Pyramid Customs (SGTX-EG-CBR-000009-5E7B) | **Role:** Clearance & Certification | **Tagline:** Declarations · Clearance · EUR.1

1-Click Actions: 1-Click File Declaration (→ declarations) | 1-Click Clear Shipment (→ clearance)

Tabs (10 total):

Group	Tab	Purpose
Overview	Command Center	Executive summary + declaration queue
Operations	Declarations (Nafeza)	File + track customs declarations via Nafeza
Operations	Clearance Status	Track clearance status + holds + releases
Operations	Physical Document Jobs	Manage physical document handling + delivery
Operations	Certificates	Issue EUR.1 + COO + other certificates
Operations	Performance	Broker performance metrics (Add-On 10)
Trade	Trade Certificates	Manage trade certificates
Finance	Invoices	Broker fee invoicing
Governance	Compliance	Broker licence + liability insurance (Add-On 10)
Governance	Audit Trail	Event history

Workflow: Receive declaration request → File declaration via Nafeza → Track clearance status → Respond to holds/queries → Confirm clearance → Issue certificates → Invoice

20.7 Financier-Bank Portal — Full Workflow

Portal ID: bank | **Tenant:** Commercial International Bank (SGTX-EG-BNK-000007-1F8D) | **Role:** Trade Finance | **Tagline:** LC · Guarantees · Settlement

1-Click Actions: 1-Click Submit Bid (→ opportunities) | 1-Click Disburse (→ portfolio)

Tabs (10 total):

Group	Tab	Purpose
Overview	Command Center	Executive summary + active financing + liquidation alerts
Operations	Opportunities	View financing requests + submit bids
Operations	Portfolio	Active financing portfolio + repayment tracking
Operations	Letters of Credit	LC management (issuance, confirmation, amendment)
Operations	Collateral	Collateral management + margin calls
Operations	Borrowers	Borrower directory + credit assessment
Operations	Proof of Reserves	Reserve attestation (Add-On 7)
Finance	Repayments	Repayment schedule + collection tracking
Governance	Compliance	AML + sanctions + regulatory reporting
Governance	Audit Trail	Event history

Workflow: Receive financing request → Assess credit + collateral → Submit bid → Trader accepts → Disburse funds → Track repayment → Settle

20.8 Financier-Private Portal — Full Workflow

Portal ID: pfi | **Tenant:** Sovereign Capital (SGTX-EG-PFI-000011-3C2E) | **Role:** Private Capital | **Tagline:** Trade Finance · Private Capital

1-Click Actions: 1-Click Submit Offer (→ opportunities) | 1-Click Release Funds (→ portfolio)

Tabs (8 total):

Group	Tab	Purpose
Overview	Command Center	Executive summary + active offers
Operations	Opportunities	View financing requests + submit offers
Operations	Portfolio	Active investment portfolio
Operations	Collateral	Collateral management
Operations	Borrowers	Borrower directory
Finance	Repayments	Repayment tracking
Governance	Compliance	Compliance + risk assessment
Governance	Audit Trail	Event history

Workflow: Receive financing request → Assess risk → Submit offer → Trader accepts → Release funds → Track repayment → Settle

20.9 Government Portal — Full Workflow (24 Tabs)

Portal ID: gov | **Tenant:** Egyptian Customs Authority (SGTX-EG-GOV-000001-9A0B) | **Role:** Customs · CBE · NFSA | **Tagline:** Regulatory Oversight · Trade Visibility

1-Click Actions: 1-Click Assess Declaration (→ customs) | 1-Click Reconcile FX (→ fx)

Tabs (28 total):

Group	Tab	Purpose
Overview	01. Command Center	Regulatory oversight dashboard
Oversight	02. National Trade Flow	National trade flow visualization
Customs	03. Customs Assessment	Assess customs declarations
Monetary	04. FX & Settlement (CBE)	Cross-border FX monitoring + reconciliation
Oversight	05. Food Safety (NFSA)	Food safety alerts + NFSA oversight
Platform	06. Integrations Health	External integrations health monitor
Governance	07. Governor Decision	View Governor decisions + audit trail
Governance	08. OPA Policies	View OPA policy registry
Governance	09. Loom Verification	Verify Loom hash chain integrity
Governance	10. Jurisdiction Matrix	Jurisdiction capability matrix (10+ jurisdictions)
Governance	11. QES Layer (Egypt Trust)	Qualified Electronic Signature oversight
Governance	12. Device Trust & Auth	Device trust + authentication audit
Governance	13. Court Evidence	Evidence packages for court proceedings
Governance	14. Compliance Screening	Compliance screening overview
Governance	15. Suspicious Activity Reports	SAR overview + FIU interface
Governance	16. USTN Master Object	USTN registry + master object view
Governance	17. Role Journey Maps	User role journey documentation
Governance	18. Audit Trail	Platform-wide audit trail
Governance	19. Integration Control Center	Phase 8 integration catalog + gap control
Governance	20. Transport & Logistics	Transport & logistics oversight
Governance	21. Financial Execution	Financial execution oversight
Governance	22. Post-Trade Completion	Post-trade completion oversight
Governance	23. Regulatory Change Center	Regulatory change management (Add-On 28 GRiRE)
Governance	24. Production Readiness	Production readiness report (INTEGRATION_REQUIRED)
Governance	GRiRE Engine	GRiRE country profile + tariff + docs lookup
Governance	Force Majeure	Force majeure event monitoring
Governance	Compliance Calendar	Regulatory deadline calendar
Governance	Regulatory Snapshots	Immutable per-trade regulatory snapshots

Workflow: Monitor trade flow → Assess declarations → Reconcile FX → Monitor food safety → Review Governor decisions → Verify Loom → Respond to SARs → Oversee production readiness

20.10 Platform Admin Portal — Full Workflow

Portal ID: admin | **Tenant:** Platform Admin (SGTX-ZZ-ADM-000001-A1B2) | **Role:** Platform Governance Authority | **Tagline:** Sovereign · Governance · Audit

1-Click Actions: 1-Click Run Audit (→ audit) | 1-Check Integrations (→ integrations)

Tabs (11 total):

Group	Tab	Purpose
Overview	Command Center	Platform-wide command center
Monitoring	Metrics & Health	Platform metrics + health dashboard
Security	Incidents	Incident management (SEV-0 through SEV-4)
Security	Threat Findings	Security threat findings (Add-On 5)
Governance	Multisig Approvals	Multisig approval queue (2-of-3 / 3-of-5)
Platform	Add-on Library	Add-on activation/deactivation
Platform	Add-Ons Hub (9-28)	Unified add-on hub with 19 sub-tabs
Platform	Competitor Benchmark	SGTX vs competitors comparison
Platform	Integrations	Integration health + management
Monitoring	SLA & Status	SLA monitoring + platform status
Governance	Governor Audit	Governor decision audit

Workflow: Monitor platform health → Manage incidents → Review multisig queue → Activate/deactivate add-ons → Check integrations → Run Governor audit

20.11 Marketplace Partner Portal — Full Workflow

Portal ID: marketplace-partner | **Tenant:** Marketplace Partner (SGTX-ZZ-MKT-000001-C3D4) | **Role:** External Platform · API Integration | **Tagline:** Leads · Webhooks · Revenue

1-Click Actions: 1-Click View Leads (→ leads) | 1-Click Generate Key (→ api-keys)

Tabs (8 total):

Group	Tab	Purpose
Overview	Command Center	Partner command center
Operations	Leads Management	View + manage inbound leads
Operations	Webhooks	Webhook configuration + logs
Finance	Revenue	Revenue sharing dashboard
Platform	API Keys	API key generation + management
Platform	Sandbox	Sandbox environment for testing
Legal	Agreement	Revenue share agreement
Admin	Company Admin	Company settings

Workflow: Receive leads → Manage webhooks → Track revenue → Generate API keys → Test in sandbox

20.12 Laboratory Portal — Full Workflow

Portal ID: lab | **Tenant:** Cairo Analytical (SGTX-EG-LAB-000014-6F4D) | **Role:** Food & Pesticide Testing | **Tagline:** Sampling · Testing · Certification

1-Click Actions: 1-Click Start Sampling (→ queue) | 1-Click Release Report (→ reports)

Tabs (9 total):

Group	Tab	Purpose
Overview	Command Center	Lab command center + test queue
Operations	Test Requests	Inbound test requests
Operations	Queue	Sampling queue management
Operations	Reports & Results	Publish test reports + certificates
Operations	Certificates	Issue lab certificates
Operations	Performance	Lab performance metrics
Finance	Invoices	Lab fee invoicing
Governance	Compliance	Lab accreditation + calibration
Governance	Audit Trail	Event history

Workflow: Receive test request → Schedule sampling → Collect sample → Conduct tests → Publish report → Issue certificate → Invoice

PART VI — TRADE LIFECYCLE

21. The 36-Stage End-to-End Trade Workflow (Art 129)

[L1] The SGTX trade lifecycle consists of 36 stages, from initial intent through final USTN closure. Each stage has defined inputs, outputs, authority class, and portal touch-points. This is the canonical trade lifecycle — every trade on the platform follows this sequence.

#	Stage	Authority	Portal	Status
1	INTENT — Trade Intent	Assertion (Buyer)	Buyer	WORKING
2	COUNTERPARTY — Known Counterparty	Assertion (Buyer)	Buyer	WORKING
3	RFQ — Request for Quote	Assertion (Buyer)	Buyer	WORKING
4	QUOTE — Quote Submitted	Assertion (Seller)	Seller	WORKING
5	NEGOTIATION — Negotiation	Assertion (Party)	Buyer/Seller	WORKING
6	PO/SO — Purchase/Sales Order	Assertion (Buyer→Seller)	Buyer/Seller	WORKING
7	PROFORMA — Proforma Invoice	Assertion (Seller)	Seller	WORKING
8	CONTRACT — Contract Locked	Confirmation (Governor)	Buyer/Seller	WORKING
9	REG_SNAPSHOT — Regulatory Snapshot	Observation (System)	Gov	WORKING
10	CLASSIFICATION — Product Classification	Assertion (System A2)	Buyer	WORKING
11	ORIGIN — Origin Determination	Assertion (System)	Buyer	WORKING
12	FTA — FTA Preference	Observation (System)	Buyer	WORKING
13	LICENSE — License Check	Observation (System)	Buyer/Seller	WORKING
14	PERMIT — Permit Check	Observation (System)	Buyer/Seller	WORKING
15	CERTIFICATE — Certificate Check	Observation (System)	CBR	WORKING
16	INSURANCE — Insurance Issued	Assertion (Insurer)	Buyer	WORKING
17	PACKING — Packing Complete	Assertion (Seller)	Seller	WORKING
18	TRANSPORT — Transport Configured	Assertion (Shipper)	Buyer/Seller	WORKING
19	BOOKING — Booking Confirmed	Assertion (Carrier)	LSP/Ship	WORKING
20	EXPORT_CUSTOMS — Export Customs	Assertion (Broker)	CBR	WORKING
21	SECURITY — Security Screening	Observation (System)	Ship	WORKING

#	Stage	Authority	Portal	Status
22	EXECUTION — Physical Execution	Observation (System)	LSP/Ship	WORKING
23	TRANSIT — In Transit	Observation (Carrier)	All	WORKING
24	IMPORT_CUSTOMS — Import Customs	Assertion (Broker)	CBR	WORKING
25	DUTY/TAX — Duty & Tax Paid	Confirmation (Bank)	Buyer	WORKING
26	INSPECTION — Inspection Complete	Assertion (QC/Lab)	QC/Lab	WORKING
27	RELEASE — Customs Release	Confirmation (Authority)	Gov	WORKING
28	DELIVERY — Delivery Confirmed	Observation (Carrier)	Buyer	WORKING
29	ACCEPTANCE — Delivery Accepted	Assertion (Receiver)	Buyer	WORKING
30	SETTLEMENT — Settlement Complete	Confirmation (Bank)	Buyer/Seller	WORKING
31	RECONCILIATION — Bank Reconciliation	Confirmation (Bank)	Bank	WORKING
32	ACCOUNTING — Accounting Complete	Assertion (System)	Admin	WORKING
33	CLAIMS — Claims/Warranty Window	Observation (System)	Buyer/Seller	WORKING
34	POST_CLEARANCE — Post-Clearance	Observation (System)	Gov	WORKING
35	EVIDENCE — Evidence Sealed	Confirmation (Governor)	Admin	WORKING
36	USTN_CLOSED — USTN Closed	Confirmation (Governor)	All	WORKING

[L1] **Automated stage triggers:** The following stages are automatically triggered by the platform:

- **Stages 1-3 (INTENT, COUNTERPARTY, RFQ):** Automatically logged when buyer submits trade request.
- **Stage 4 (QUOTE):** Automatically logged when seller submits quote.
- **Stage 8 (CONTRACT):** Automatically logged when contract is locked + USTN minted.
- **Stage 9 (REG_SNAPSHOT):** Automatically captured when contract is locked (regulatory snapshot with SHA-256 hash).
- **Stages 10-18:** Automatically evaluated during contract lock (classification, origin, FTA, license, permit, certificate, insurance, packing, transport).
- **Stage 35 (EVIDENCE):** Automatically assembled when all 7 closure conditions are met.
- **Stage 36 (USTN_CLOSED):** Automatically set when evidence is sealed + multisig approval.

22. Trade Initiation Flow (Buyer→Seller)

[L1] The trade initiation flow describes what happens when a buyer submits a trade request:

- **Step 1:** Buyer clicks '1-Click New Trade' on the Buyer Command Center.
- **Step 2:** 11-step wizard opens. Buyer fills: seller, incoterm, commodity, HS code, containers, documentation, transport, insurance, settlement, criticality, shipments.
- **Step 3:** Governor pre-screens the trade (G1-G7 gates).
- **Step 4:** Compliance screening (sanctions, KYB) runs synchronously.
- **Step 5:** Trade is created with status PENDING_SELLER_RESPONSE + temporary USTN (SGTX-PEND-...).
- **Step 6:** Seller receives Smart Inbox notification (priority 75, 'New trade request').
- **Step 7:** Buyer receives confirmation inbox item (priority 70, 'Trade request initiated').
- **Step 8:** Government receives REGULATORY_OVERSIGHT inbox item (priority 60, 'New trade initiated').
- **Step 9:** Brain event 'trade.created' published to 38+ downstream subscribers.
- **Step 10:** Trade Stage Log records: INTENT, COUNTERPARTY, RFQ stages completed.
- **Step 11:** Contacts auto-saved to both parties' networks (relationship-controlled, not marketplace).

23. Quote & Negotiation Flow

[L1] The quote and negotiation flow describes what happens when a seller submits a quote:

- **Step 1:** Seller clicks '1-Click Submit Quote' on the Seller Command Center.
- **Step 2:** Quote builder opens. Seller sets: EXW price, packing plan, commodity line items, SGTX fee (1.5%).
- **Step 3:** Governor evaluates the quote (transactional — all mutations in db.$transaction).
- **Step 4:** Trade status updated to QUOTED (atomically with inbox creation).
- **Step 5:** Buyer receives Smart Inbox notification (priority 75, 'Quote received').
- **Step 6:** Government receives REGULATORY_OVERSIGHT inbox item (priority 55, 'Quote submitted').
- **Step 7:** Brain event 'trade.quote.submitted' published.
- **Step 8:** Trade Stage Log records: QUOTE stage completed.
- **Step 9:** If buyer counters → negotiation round created → NEGOTIATION stage logged.
- **Step 10:** If buyer accepts → PO/SO creation begins → PO/SO stage logged.

24. Contract Lock & USTN Minting

[L1] The contract lock flow is the most critical stage — it mints the USTN and captures the regulatory snapshot:

- **Step 1:** Both parties sign the contract + addenda.
- **Step 2:** Governor verifies 4 lock conditions: (a) buyer signed, (b) seller signed, (c) fee locked, (d) release acknowledged.

- **Step 3:** USTN is minted: SGTX-{BUYER6}-{SELLER6}-{YYYYMMDDHHMMSS}-{RAND8}.
- **Step 4:** Trade status updated to CONTRACT_SIGNED, phase set to 3.
- **Step 5:** All shipments updated with the minted USTN.
- **Step 6:** Regulatory Snapshot captured: origin/destination/HS/tariff/sanctions/FTA/licenses/permits/certificates → SHA-256 hash computed → stored immutably.
- **Step 7:** Trade Stage Log records: CONTRACT + REG_SNAPSHOT stages completed.
- **Step 8:** Both parties notified (priority 75, 'Contract locked').
- **Step 9:** Timeline event created (phase 3 complete).

25. Settlement & Closure Flow

[L1] The settlement and closure flow is the final stage of the trade lifecycle:

- **Step 1:** Delivery accepted by receiver (POD signed) → condition 1 met.
- **Step 2:** All payment legs settled (bank confirms) → condition 2 met.
- **Step 3:** Bank reconciliation matches SGTX records → condition 3 met.
- **Step 4:** Customs complete (import + export cleared) → condition 4 met.
- **Step 5:** Post-clearance complete (audit or 'no audit required') → condition 5 met.
- **Step 6:** All disputes resolved or time-barred → condition 6 met.
- **Step 7:** Evidence package assembled (26 categories) + SHA-256 hash + sealed → condition 7 met.
- **Step 8:** canClose predicate returns true (all 7 conditions met).
- **Step 9:** Multisig approval (2-of-3) for USTN closure.
- **Step 10:** USTN marked CLOSED → evidence sealed → post-closure observation begins (90 days).
- **Step 11:** Trade Stage Log records: EVIDENCE + USTN_CLOSED stages completed.
- **Step 12:** All parties notified (priority 80, 'USTN closed').

PART VII — APPENDICES

Appendix A — Financial Control Framework (CFO)

A.1 Fee Schedule

[L2] **SGTX Platform Fee:** 1.5% of trade value, collected at settlement via the bank (non-custodial). The fee is locked at trade initiation (FeeLock) and collected by the bank during settlement — SGTX never receives the fee directly.

[L2] **Optional service fees:** QC inspection (buyer-requested), lab tests (buyer-requested), customs broker services, logistics RFQ fees. These are quoted by the service provider and accepted by the trader; SGTX does not set these prices (non-marketplace principle).

A.2 Reserve Policy

[L0-17] **110% reserve rule:** If reserve metadata is maintained, it must be at least 110% backed. The constitutional layer sets the threshold; ZK attestation (Add-On 7) provides cryptographic evidence.

[L2] **Composition:** Reserves may consist of cash, bank guarantees, government bonds, or other approved instruments. The composition is jurisdiction-aware.

A.3 Non-Custody Attestation Template

'SGTX Platform does not hold, receive, or transfer customer funds at any point in the trade lifecycle. All funds flow through connected banks and regulated payment service providers. SGTX orchestrates payment instructions but never becomes the settlement authority. The FeeLock mechanism locks the platform fee at trade initiation, but the fee is collected by the bank during settlement.'

A.4 Unit Economics

[L2] **Revenue:** 1.5% fee × trade volume. Example: $100,000 trade = $1,500 SGTX fee.

[L2] **Cost structure:** Infrastructure (Turso, Vercel, AI providers), personnel, compliance, legal. Variable cost per trade: ~$2-5 (AI inference + DB + API). Contribution margin: ~97% at scale.

Appendix B — Regulatory & Legal

B.1 Regulatory Classification Matrix (Egypt)

Functionality	Egypt Classification	Required Licence	Status
Trade orchestration	Service provider	None (non-regulated)	OPERATIONAL
Payment instruction	Payment service provider	CBE registration (if applicable)	LEGAL_AUTHORIZATION_REQUIRED
Customs declaration	Customs broker facilitation	Broker licence (broker-side)	OPERATIONAL
Document issuance	Digital document service	ETA/CargoX integration	CORE_READY
Settlement orchestration	Non-custodial orchestration	Bank partnership	LEGAL_AUTHORIZATION_REQUIRED
AI advisory	AI service provider	None (advisory only)	OPERATIONAL

B.2 Data-Residency Element Classification

[L2] EGYPT_ONLY (must stay in Egypt), COUNTRY_ONLY, REGIONAL, APPROVED_CROSS_BORDER, GLOBAL_ALLOWED. Strictest applicable rule wins.

B.3 SAR/FIU Interface Notes

[L2] SGTX provides structured evidence packages for SAR filing. The platform does not file SARs directly (bank's responsibility). Add-On 17 and Add-On 7 support the evidence chain.

Appendix C — Operating Model

C.1 Platform Governance Authority RACI

Decision	Responsible	Accountable	Consulted	Informed
Constitutional amendment	Governance Authority	Signatories (3-of-5)	Legal, Compliance, Eng	All
Policy update (OPA)	Policy Team	Governance Authority	Eng, Compliance	Operations
WASM module update	Engineering	CTO	Security, Governance	Operations
Integration activation	Integration Team	CTO	Legal, Compliance	Operations
Incident (SEV-0)	On-call Engineer	CTO	Security, Legal, Comms	Board
Incident (SEV-1)	On-call Engineer	Eng Lead	Security	Operations

C.2 Multisig Policy

[L2] **Standard:** 2-of-3 for trade-level irreversible actions (USTN closure, contract lock, settlement release). **Constitutional:** 3-of-5 for platform-level changes. QES required.

C.3 Change-Management Process

[L2] 4-stage: Author → Review (security + legal) → Approve (multisig 3-of-5) → Deploy (staged: sandbox → production). Each stage Loom-logged.

C.4 Incident Severity Model

Severity	Definition	Response	Escalation
SEV-0	Platform-wide outage / data breach	Immediate (24/7)	CTO + Legal + Comms + Board
SEV-1	Major feature failure (multiple trades)	1 hour	Eng Lead + CTO
SEV-2	Feature failure (single trade/tenant)	4 hours	On-call Engineer
SEV-3	Minor bug with workaround	24 hours	Assigned Engineer
SEV-4	Cosmetic / enhancement	Next sprint	Product Backlog

Appendix D — Implementation Priority Framework

[L2] P0–P4 waves are dependency-driven, not calendar-driven. No contractual calendar claims.

Wave	Scope	Dependencies	Status
P0 — Core Trade Execution	Trade, Contract, USTN, Governor, Event Spine, State Vector, Closure Policy	None (foundation)	CORE_READY
P1 — Transport Engines	Road, Air, Ocean Container, RoRo, Rail, Multimodal Orchestrator	P0 complete	CORE_READY
P2 — Intelligence & Compliance (Add-Ons 1-8)	GNN, Federated Learning, Causal, Self-Healing, Pentest, PQC, ZK, Customs Bond	P0 complete	CORE_READY
P3 — Operational Add-Ons (9-18)	Demurrage, Broker Liability, Valuation, Cold Chain, Inspection, Currency, Sandbox, FTA, Security, Calendar	P0 + GRiRE	CORE_READY
P4 — Extended (19-28)	Cargo Insurance, Trade Finance, Back-to-Back LC, Force Majeure, Export Docs, Port/Terminal, Payment Guarantee, Demurrage Dispute, GRiRE	P0 + P2	CORE_READY

Appendix E — Full Historical Audit Trail (Immutable Annex)

This annex preserves the full audit trail from v13.1 Part A (findings A-01 through A-24). Nothing has been removed. The audit findings are the governing resolutions for all v14.0 architecture.

Finding	Conflict	Resolution (Governing)
A-01	Direct Government API Scope	Four Egypt connectors are initial reference; worldwide adapter fabric is extensibility layer.
A-02	AI A4 Authority Language	A4 is deterministic policy automation; AI never has independent execution authority.
A-03	Non-Custody vs Payment	Non-custody is architectural; Bank Settlement Gateway orchestrates but never holds funds.
A-04	Stablecoin/DeFi vs Bank Settlement	DeFi is conditional sub-rail; bank-authoritative settlement is canonical.
A-05	Reserve Metadata vs Custody	Reserve tables store attestations only; do not create custody.
A-06	Reserve Ratio Thresholds	110% rule is constitutional; ZK attestation provides evidence.
A-07	GNN Non-Marketplace	Trust graph for known parties; never discovers/recommends/ranks.
A-08	Seven vs Twenty-Eight Add-Ons	Full 28-add-on catalogue is canonical; original seven are historical reference.
A-09	RoRo as First-Class	RoRo is first-class transport mode, not ocean sub-mode.
A-10	Egypt Nafeza Mode-Specific	Mode-specific applicability; not one generic maritime workflow.
A-11	Closure Semantics	7-condition closure; CLOSED_WITH_EXCEPTION for severity ≤2 only.
A-12	USTN as Namespace	Universal reference; does not override external authorities.
A-13	ISO 20022 as Output	Integration output format, not canonical data.
A-14	Egypt Data Localisation	EGYPT_ONLY for specific elements; strictest rule wins.
A-15	Zero-Cost Clarified	Institutional-cost scope: data free; institutional costs real.
A-16	Marketplace Terminology	Relationship-controlled orchestration, NOT marketplace.
A-17	AI Explanations vs Decisions	AI generates explanations; Governor makes authoritative decisions.
A-18	Manual Fallback	Governed: authenticated, attributable, timestamped, Loom-logged.
A-19	Global Readiness Claims	Use CORE_READY/PRODUCTION_CONNECTED; never WORLDWIDE_INTEGRATED.
A-20	Command vs Event	Command (intent) ≠ Event (fact); Commands to Governor, Events to Spine.
A-21	Recovery vs Rollback	Recovery restores state; never erases history. Loom immutable.
A-22	Proof-of-Reserves Wallet Examples	Verification-only; consistent with non-custody.
A-23	Jurisdiction Examples	Illustrative configuration governed by active jurisdiction profile.
A-24	Incomplete Change-Set	Add-Ons 9-28 and Final Summary now fully integrated.

Appendix F — Source Manifest & SHA-256 Hashes

Source	Type	Lines	SHA-256
SGTX_PLATFORM_MASTER_BLUEPRINT_INTEGRATED_v12.0.docx	Main Blueprint	~76,122	c263f527f3966dab01d3cffc87e7d2747d01c017ca7a786d1964f09018087d42
sgtx add ons and modifications.rtf	Change-Set	~7,582	87181d220df82a485485eec4b9896031910c79afa6332516ef2afac4682c5b72
SGTX_v13.1_FINAL.docx	Integrated Edition	~47,216	(computed from extracted text)

Appendix G — Change Log v13.1 → v14.0

Change	Rationale
Three-layer separation (L0/L1/L2)	Separate immutable principles from mutable architecture from testable specs
Over-claim elimination	'production-ready' → CORE_READY; 'complete' → 'specified'; 'zero-cost' → 'institutional-cost scope'
28-add-on status matrix	Explicit deployment-state vocabulary for every add-on
4-dimension external readiness	TECHNICAL/LEGAL/OPERATIONAL/COMMERCIAL reported independently
Command/Event taxonomy formalized	Command (intent) ≠ Event (fact) — enforced in API contract
32-point constitution	All 32 constitutional points listed explicitly with [L0] tags
Audit trail preserved as annex	A-01 through A-24 preserved as governing resolutions
RTO/RPO with quorum definitions	Explicit durability/quorum requirements per tier
No calendar claims	P0–P4 waves are dependency-driven, not date-driven
Full portal documentation (12 portals)	Each portal's tabs, screens, workflows documented in full
36-stage trade lifecycle	All 36 stages with authority class, portal, status documented
Automated stage triggers	Stages 1-3, 4, 8-9, 35-36 auto-triggered by platform
Quote model	Dedicated Quote Prisma model (replaces JSON-in-Trade.globalNotes)
Multi-model AI consensus	3 providers (Gemini+Groq+HuggingFace) with weighted voting
Smart Inbox Priority	AI-computed priority: basePriority×0.4 + tradeValue×0.3 + urgency×0.2 + criticality×0.1
Evidence Package download	26-category JSON evidence package at USTN closure
Trade Cost Calculator	True Landed Cost (18 components per Art 24)
Regulatory Pre-Check	Pre-submit sanctions+FTA+docs+duty+cold chain check
Competitor Benchmark	SGTX vs TradeLens/Maersk Spot/Flexport/CargoX comparison
SSE Realtime Notifications	Server-Sent Events for live inbox + trade updates
1-Click Action Bar	Every portal dashboard has 1-Click Trade + 1-Click Payment buttons
1-Click Close USTN	7-condition checklist + closure ceremony on buyer/gov portals
Trade Lifecycle Visualizer	36-stage progress bar on every Command Center

Appendix H — Dependency Graph of Major Components

[L1] The following dependency graph shows how major SGTX components relate to each other:

Layer 0 (Constitution) → **Layer 1 (Architecture)** → **Layer 2 (Implementation)**

Governor Pipeline: Governor → OPA → WasmEdge → Loom

- Governor receives Commands from all components
- OPA evaluates policies (sidecar)
- WasmEdge executes constitutional rules (deterministic)
- Loom logs all decisions (hash-chained, immutable)

Event Flow: Command → Governor → Event Spine → State Vector → Transaction Twin → Closure Policy

- Command enters Governor pipeline
- If ALLOW, Event appended to Event Spine (hash-chained)
- Event updates State Vector (12 domain clocks)
- Transaction Twin mirrors state for post-closure observation
- Closure Policy evaluates 7 conditions for USTN closure

Settlement Flow: Settlement Orchestration → Bank Settlement Gateway → Bank → Payment Settled Event

- Settlement Orchestration creates multi-leg instructions
- Bank Settlement Gateway validates (6-stage pipeline)
- Bank executes settlement (non-custodial — SGTX never holds funds)
- Bank emits Payment Settled Event → Event Spine

Transport Flow: Multimodal Orchestrator → Road/Air/Ocean/RoRo/Rail Engines → Terminal Adapters

- Multimodal Orchestrator coordinates cross-mode handoffs
- Each mode engine owns mode-specific rules
- Terminal adapters interface with port/airport/border systems

AI Flow: AI Subsystem (A1-A3) → Governor → A4 Execution → Event Spine

- AI produces recommendation (A1 advisory, A2 constraining, A3 escalation)
- Recommendation submitted to Governor as Command
- Governor evaluates (G1-G7, OPA, WasmEdge)
- If ALLOW, A4 executes deterministically (NOT AI autonomy)
- Event appended to Spine + decision logged to Loom